HBA Lecture Notes in Mathematics
IMSc Lecture Notes in Mathematics

Robert Tubbs

Hilbert's Seventh Problem

Solutions and Extensions

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Contents

Preface vii

1 Hilbert's seventh problem: Its statement and origins 1

2 The transcendence of e, π and $e^{\sqrt{2}}$ 13

3 Three partial solutions 21

4 Gelfond's solution 33

5 Schneider's solution 45

6 Hilbert's seventh problem and transcendental functions 61

7 Variants and generalizations 73

References 81

Bibliography 81

Index 83

Preface

The twenty-three problems David Hilbert posed at the Second International Congress of Mathematicians in 1900 proved to inspire many of the mathematical breakthroughs of the twentieth, and nascent twenty-first, centuries. Hilbert's seventh problem, whose solutions are the topic of this monograph, concerned the transcendence of numbers either of a particular form or that are given as special values of transcendental functions. In this short report we will look first at the results in transcendence theory that preceded Hilbert's lecture. This brief introduction to the earliest results in transcendence theory is followed by a dissection of Hilbert's statement of his seventh problem. We will then look at three partial solutions that were given some thirty years later. These partial solutions were soon followed by two solutions to the most commonly cited portion of the seventh problem. We will then look at some early progress on another aspect of Hilbert's problem and finally look at a particularly interesting late-twentieth century advance.

This monograph grew out of some notes I prepared for students attending my short course *Hilbert's Seventh Problem: Its solutions and extensions* at the Institute for Mathematical Sciences in Chennai, India in December 2010. It is written for students and faculty who want to explore the progression of mathematical ideas that led to the partial solutions then complete solutions to one portion of Hilbert's problem. I thank the gifted students who attended my lectures for their many comments, questions, and corrections. This text owes a great deal to them. Of course I am solely responsible for any errors that remain.

I thank Professor Sanoli Gun of the IMSc for inviting me to participate in the institute's Number Theory Year dedicated to the institute's talented director Professor R. Balasubramanian. I also thank Professor Gun and Professor Purusottam Rath of the Chennai Mathematical Institute for the hospitality they extended to me during my stay in Chennai. I also want to acknowledge the referee who made several thoughtful comments. Lastly I thank Professor Michel Waldschmidt of Paris for suggesting that I participate in the institute's stimulating program.

About the Author

Robert Tubbs is Associate Professor of Mathematics at the University of Colorado Boulder, United States. His research interest lies in number theory, especially transcendental number theory, the intellectual history of mathematical ideas and mathematics and the humanities.

Chapter 1

Hilbert's seventh problem: Its statement and origins

At the second International Congress of Mathematicians in Paris, in 1900, the mathematician David Hilbert was invited to deliver a keynote address, just as Henri Poincaré had been invited to do at the first International Congress of Mathematicians in Zurich in 1896.

According to a published version of Hilbert's lecture [14], which appeared soon thereafter, he began his lecture with a bit of a motivation for offering a list of problems to inspire mathematical research:

...the close of a great epoch not only invites us to look back into the past but also directs our thoughts to the unknown future. The deep significance of certain problems for the advance of mathematical science in general and the important role which they play in the work of the individual investigator are not to be denied. As long as a branch of science offers an abundance of problems, so long is it alive; a lack of problems foreshadows extinction or the cessation of independent development. Just as every human undertaking pursues certain objects, so also mathematical research requires its problems.

In his lecture Hilbert posed ten problems. In the published versions of his lecture Hilbert offered twenty-three problems (only eighteen could really be considered to be problems rather than areas for further research). The distribution of his published problems is, roughly: two in logic, three in geometry, seven in number theory, ten in analysis/geometry, and one in physics (and its foundations). To date sixteen of these problems have been either solved or given counterexamples.

What will concern us in these notes is the seventh problem on Hilbert's list, which concerns the arithmetic nature of certain numbers–in particular, Hilbert proposed that certain, specific, numbers are transcendental, i.e., not algebraic and so not the solution of any integral polynomial equation $P(X) = 0$.

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Some relevant early developments

By the time Hilbert spoke in Paris, transcendental number theory had already had a short yet fairly glorious history. The beginning of this history, so far as we can tell from what was written down, began with either L. Euler or G. Leibniz. These two mathematicians must have felt a certain exasperation when they sensed that some numbers were just beyond their grasps; these *transcendental* numbers were not numbers that one would ordinarily encounter through algebraic methods. Rather than trace this history here we highlight research that influenced the early development of the study of transcendental numbers and then the nineteenth-century developments that inspired Hilbert's seventh problem.

A natural place to begin is with Euler. In 1748 Euler published *Introductio in analysin infinitorum* (Introduction to Analysis of the Infinite) [4] in which he did several things that are relevant to the development of transcendental number theory and to these notes:

• Euler summed the series $1/1^k + 1/2^k + 1/3^k + \ldots$ for all even $k, 2 \leq k \leq 26$, which we represent by $\zeta(k)$. (Euler showed that $\zeta(2) = \frac{\pi^2}{6}$, [1].)

• Euler derived the formula

$$e^{ix} = \cos(x) + i\sin(x).$$

• From the above equality, Euler obtained the relationship

$$e^{i\pi} = -1.$$

• Finally, Euler found continued fraction expansions for e and e^2 (which were non-terminating thus showing that each of these numbers is irrational).

In his book Euler also made an interesting conjecture concerning the nature of certain logarithms of rational numbers. As Euler's conjecture remarkably foreshadows one part of Hilbert's seventh problem, it is worth stating. Euler wrote:

... the logarithms of [rational] numbers which are not the powers of the base are neither rational nor irrational ...

When Euler used the terminology "irrational" he meant what we would call "irrational and algebraic." Euler went on to say that *... it is with justice that [the above logarithms] are called transcendental quantities.*

Euler's conjecture has the simple, more modern, formulation:

Conjecture (Euler). *For any two positive rational numbers $r \neq 1$ and s, the number*

$$log_r s \quad \left(= \frac{\log s}{\log r} \right)$$

is either rational or transcendental.

This conjecture applies to each of the numbers $\log_{\frac{1}{2}} 4 = -2, \log_{\frac{1}{2}} 3$ and $\log_3 5$, asserting that the latter two are transcendental. But it also asserts that a number like $2^{\sqrt{2}}$ is irrational, because it says that a rational number to an irrational, algebraic power cannot be rational.

Although Euler had speculated that there might exist non-algebraic numbers, none were known, indeed they were not even known to exist until the work of Liouville in the middle of the nineteenth century (when Liouville proved that transcendental numbers exist by exhibiting one). This result is actually the corollary of a theorem Liouville established concerning *how well* an algebraic number can be approximated by a rational number:

Theorem 1.1 (Liouville, 1844, [22]). *Suppose α is an algebraic number of degree $d > 1$. Then there exists a positive constant $c(\alpha)$ such that for any rational number $\dfrac{a}{b}$*

$$\left|\alpha - \frac{a}{b}\right| > \frac{c(\alpha)}{b^d}. \tag{1.1}$$

The proof of Liouville's Theorem is entirely elementary, requiring only an application of the Mean Value Theorem from calculus and a bit of cleverness. Because this proof is so straightforward it is worth seeing it at least once (it is outlined in the exercises at the end of this chapter).

It is an elementary exercise to deduce that the number ℓ, below, does not satisfy the conclusion of Liouville's Theorem for any constant $c(\ell)$ or for any degree d :

$$\begin{aligned}\ell = \sum_{n=1}^{\infty} 10^{-n!} &= \frac{1}{10} + \frac{1}{100} + \frac{1}{1000000} + \frac{1}{100000000000000000000000} + \cdots \\ &= .1\ 1\ \underbrace{000}_{3\text{ zeros}}\ 1\ \underbrace{00000000000000000}_{17\text{ zeros}}\ 1\ \underbrace{000\ldots00}_{95\text{ zeros}}\ 1\ 000\ldots\end{aligned}$$

Although Euler's discovery that e has a non-repeating continued fraction expansion implies that e is irrational, this discovery is not in the current of mathematical thought that includes Hilbert's seventh problem. That honor belongs to a proof of the irrationality of e that was published by Joseph Fourier in 1815 [29]. Fourier's proof depends on the series representation

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

(We will look at this instructive proof below.)

Then in 1873 Hermite [13] established the transcendence of e, see next chapter, by using the series representation for the numbers

$$e^n = \sum_{k=0}^{\infty} \frac{n^k}{k!}.$$

Just under a decade later, in 1882, Lindemann [21] used the series representation for e^z and results about algebraic numbers to establish that for any nonzero algebraic number α, the number e^α is transcendental. (The transcendence of $i\pi$ and so of π follows since, by Euler's formula, $e^{i\pi} = -1$.)

Two years later, in 1884, Weierstrass [33] supplied the proof of something Lindemann had claimed, but not established. This is now called the Lindemann-Weierstrass Theorem.

Theorem 1.2 (Lindemann-Weierstrass). *Suppose $\alpha_1, \alpha_2, \ldots, \alpha_\ell$ are distinct, nonzero algebraic numbers. Then any number of the form*

$$a_1 e^{\alpha_1} + a_2 e^{\alpha_2} + \cdots + a_\ell e^{\alpha_\ell},$$

with all a_j algebraic and not all zero, is transcendental.

Back to Hilbert's Lecture

With the above survey as background we are now in a better position to understand the scope of Hilbert's proposed seventh problem. Hilbert began his statement of this problem with:

Hermite's arithmetical theorems on the exponential function and their extension by Lindemann are certain of the admiration of all generations of mathematicians.

Following his introductory words Hilbert continued:

I should like, therefore, to sketch a class of problems which, in my opinion, should be attacked as here next in order.

We will see that indeed Hilbert's seventh problem is not a single problem, but has separate parts. One of these concerns the values of certain functions:

... we expect transcendental functions to assume, in general, transcendental values for ... algebraic arguments ... we shall still consider it highly probable that the exponential function $e^{i\pi z}$ ... will ... always take transcendental values for irrational algebraic values of the argument z

We recall that a function $f(z)$ is a transcendental function if there does not exist a nonzero polynomial $P(x, y)$ so that the function $P(z, f(z))$ is identically zero. For example e^z is a transcendental functions whereas $\sqrt{z}$ and $z^5 - z^2 + 1$ are not. So Hilbert speculated that if $f(z)$ is a transcendental function and α is a nonzero algebraic number, then $f(\alpha)$ should be a transcendental number. By what we have already seen, the only example Hilbert knew was the transcendental function $f(z) = e^z$, which by the Lindemann-Weierstrass Theorem is transcendental whenever $z = \alpha$ is a nonzero, algebraic number α.

After referring to a geometric version of his conjecture (see exercises) Hilbert went to another part of his seventh problem, stating that he believed:

the expression α^β, for an algebraic base and an irrational algebraic exponent, e. g., the number $2^{\sqrt{2}}$ or e^π, always represents a transcendental or at least an irrational number.

Hilbert's suggestion is that α^β should be transcendental, or at least irrational, whenever it has an algebraic base α (which implicitly requires $\alpha \neq 0, 1$) and an irrational algebraic exponent β. It is interesting to compare Hilbert's conjecture with Euler's conjecture from a century and a half earlier. If we slightly reformulate Euler's conjecture we will see that this portion of Hilbert's seventh problem simply expanded the arithmetic nature of the numbers under consideration.

Euler's Conjecture (1748). *If a is a nonzero rational number and β is an irrational algebraic number then a^β is irrational.*

This conjecture asserts, for example, that $2^{\sqrt{2}}$ is irrational.

Hilbert's Conjecture (1900). *If α and β are algebraic, with $\alpha \neq 0$ or 1, and β irrational then α^β is transcendental.*

This conjecture asserts that $2^{\sqrt{2}}, 2^{\sqrt{-2}}, i^{\sqrt{2}}$ and $e^\pi (= (-1)^{-i})$ are all transcendental.

Just as Euler's conjecture can be stated in terms of either exponents or logarithms, Hilbert's conjecture concerning α^β has alternate formulations. These equivalent formulations have played important roles not only in the eventual solutions of Hilbert's seventh problem in the early 1930s, but to questions that have become central to transcendental number theory. Rather than return to these equivalent statements later it seems best to give them now. (It is an interesting exercise, and it is at the end of this chapter, to establish the equivalencies of these three versions of Hilbert's conjecture.)

Second Version of Hilbert's Conjecture. *Suppose ℓ and β are complex numbers, with $\ell \neq 0$ and β irrational. Then at least one of the numbers*

$$\beta,\ e^\ell,\ e^{\beta\ell}$$

is transcendental.

Third Version of Hilbert's Conjecture. *Suppose α and β are nonzero algebraic numbers. If $\dfrac{\log \alpha}{\log \beta}$ is irrational then it is transcendental.*

Notice that this third version of Hilbert's conjecture is Euler's conjecture concerning the ratio of logarithms where the rational numbers are replaced by algebraic numbers.

This part of Hilbert's seventh problem, i.e., the transcendence of α^β, was solved independently by A. O. Gelfond and Th. Schneider, in 1934, using similar methods. In order to appreciate their solutions to this problem, and how their methods extend to other problems, it is useful to first understand earlier developments. The ones that will aid our understanding of twentieth-century

transcendental number theory most are Fourier's proof for the irrationality of e (1815), Hermite's proof for the transcendence of e (1873), and Lindemann's proof for the transcendence of π (1882).

We begin with Fourier's simple proof [29] of the irrationality of e, which we reorganize to suit our purposes.

Theorem 1.3. *e is irrational.*

As we have already noted, at the heart of Fourier's proof is the simplicity, and regularity, of the power series representation for e

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}.$$

Proof. Suppose $e = \frac{B}{A}$ where A and B are positive integers. The presumed rationality of e translates to the relationship $Ae - B = 0$. Fourier's idea is to replace e by its series representation and then truncate the series into a main term and a tail (knowing that the tail may be made as small as desired). This idea will eventually yield a positive integer that is strictly less than 1.

If we substitute the power series representation for e into the above equation we obtain the equation:

$$A\Big(\sum_{k=0}^{\infty} \frac{1}{k!}\Big) - B = 0. \tag{1.2}$$

For any integer $N \geq 1$ it is possible to separate the power series for e into a *main term*, M_N, and *tail*, T_N,

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} = \underbrace{\sum_{k=0}^{N} \frac{1}{k!}}_{M_N} + \underbrace{\sum_{k=N+1}^{\infty} \frac{1}{k!}}_{T_N}.$$

If we substitute this expression into (1.2) we have an equation:

$$A\Big(M_N + T_N\Big) - B = 0. \tag{1.3}$$

Fourier's idea is to rewrite this equation as $A \times M_N - B = -A \times T_N$ and, realizing that $|T_N|$ may be made to be a small quantity, obtain an inequality of the form

$$|\text{ nonzero integer }| < \text{a small quantity.}$$

It is easiest to follow this argument if we view (1.3) as:

$$A\Big(\underbrace{\sum_{k=0}^{N} \frac{1}{k!}}_{M_N} + \underbrace{\sum_{k=N+1}^{\infty} \frac{1}{k!}}_{T_N}\Big) - B = 0.$$

which, after using $N!$ for a common denominator for the fractions in the main term, may be rewritten as:

$$A\Big(\sum_{k=0}^{N}\frac{\left(\frac{N!}{k!}\right)}{N!}+\sum_{k=N+1}^{\infty}\frac{1}{k!}\Big)-B=A\Big(\frac{1}{N!}\big(\underbrace{\sum_{k=0}^{N}\frac{N!}{k!}}_{M_N^*}\big)+\underbrace{\sum_{k=N+1}^{\infty}\frac{1}{k!}}_{T_N}\Big)-B=0.$$

Note that for each $k,\ 0\le k\le N$, the fraction $\dfrac{N!}{k!}$ in the modified main term, M_N^*, is a positive integer, thus so is their sum. For clarity we rewrite the above equation as:

$$A\left(\frac{1}{N!}\times M_N^*+T_N\right)-B=0.$$

If we multiply this equation by $N!$ and rearrange terms slightly, we obtain:

$$|A\times M_N^*-N!B|=N!\times A\times T_N$$

The remainder of the proof requires two parts:

Part 1. Show that the expression $A\times M_N^*-N!\times B$ is a nonzero integer.

Part 2. Show that for a suitably chosen $N,\ N!\times A\times T_N$ is less than 1.

If we establish Part 1 and Part 2 we will have the conclusion:

$$0<|A\times M_N^*-N!\times B|<1,$$

where the expression $A\times M_N^*-N!\times B$ is an integer. This is, of course, a contradiction.

Establishing Part 1. Since the main term is a truncation of the series representation for e, and we are assuming e is rational and equals $\frac{B}{A}$, if $A\times M_N^*-N!\times B=0$ we obtain the contradictory inequalities:

$$e=\frac{B}{A}=\frac{M_N^*}{N!}=M_N<e.$$

Note that this holds for any N.

Establishing Part 2. We have:

$$\begin{aligned}N!\times A\times T_N&=N!\times A\sum_{k=N+1}^{\infty}\frac{1}{k!}=A\sum_{k=N+1}^{\infty}\frac{N!}{k!}\\&=A\left(\frac{1}{N+1}+\frac{1}{(N+1)(N+2)}+\frac{1}{(N+1)(N+2)(N+3)}+\cdots\right)\end{aligned}$$

We are free to specify a value for N; taking $N+1 = 2A$ the above sum becomes

$$\begin{aligned} &= \frac{A}{2A} + \frac{A}{(2A)(2A+1)} + \frac{A}{(2A)(2A+1)(2A+2)} + \cdots \\ &< \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots \\ &= 1\ . \end{aligned}$$

Thus we have deduced that the positive integer $|A \times M_N^* - N! \times B|$ is less than 1. This contradiction establishes the irrationality of e.

Rather than leaving this proof behind, its outline is so important that it deserves to be summarized. This proof consists of a sequence of easily understood steps. The proof begins with the assumption that e is a rational number. This assumption, followed by a simple argument using the power series for e, leads to a nonzero, positive integer that is less than 1. We will see that this basic structure holds in many, indeed almost all, transcendence proofs. And almost always, the most difficult part of the proof is to show that the integer derived in the proof is not equal to zero.

Before we explore these difficulties let's look at an instructive failed proof: an attempt to establish the transcendence of e through a direct application of Fourier's approach.

Sketch of the proof

Assume e is algebraic, so we have an integral polynomial equation $P(e) = 0$, explicitly, there exist integers $r_0, \ldots, r_d$, not all zero, so that

$$r_0 + r_1 e + r_2 e^2 + \cdots + r_d e^d = 0.$$

Generalizing Fourier's method, for each $n, 1 \le n \le d$, use the series representation:

$$e^n = \sum_{k=0}^{\infty} \frac{n^k}{k!} = \underbrace{\sum_{k=0}^{N} \frac{n^k}{k!}}_{M_N(n)} + \underbrace{\sum_{k=N+1}^{\infty} \frac{n^k}{k!}}_{T_N(n)}$$

Substituting each of these expressions into the presumed vanishing algebraic relationship above, we obtain:

$$r_0 + r_1\big(M_N(1) + T_N(1)\big) + r_2\big(M_N(2) + T_N(2)\big) + \cdots + r_d\big(M_N(d) + T_N(d)\big) = 0,$$

which yields:

$$\begin{aligned} &\big|r_0 + r_1 M_N(1) + r_2 M_N(2) + \cdots + r_d M_N(d)\big| \\ &\qquad = \big|r_1 T_N(1) + r_2 T_N(2) + \cdots + r_d T_N(d)\big| \end{aligned} \quad (1.4)$$

We rewrite each term $M_N(n)$ as

$$M_N(n) = \frac{1}{N!}\Big(\underbrace{\sum_{k=0}^{N} \frac{N!}{k!} n^k}_{M_N^*(n)}\Big),$$

where we note that $M_N^*(n)$ is an integer. So if we multiply (1.4) by $N!$ we obtain:

$$\begin{aligned}\big|N! r_0 + r_1 M_N^*(1) + r_2 M_N^*(2) + \cdots + r_d M_N^*(d)\big| \\ = N! \big|r_1 T_N(1) + r_2 T_N(2) + \cdots + r_d T_N(d)\big|\end{aligned}$$

Each of the terms $T_N(n)$ is of the form

$$\frac{n^{N+1}}{(N+1)!} \times \text{a convergent series;}$$

indeed

$$\big|T_N(n)\big| < \frac{n^{N+1}}{(N+1)!} \times e^n.$$

From these inequalities it is possible to obtain:

$$0 < |\text{complicated nonzero integer}| \quad \le N! \frac{d^{N+1}}{(N+1)!} \times \underbrace{e^d \times d \max\{|r_1|, \ldots, |r_d|\}}_{\text{a fixed quantity}}.$$

Unfortunately, as $N \to \infty$ the right-hand side of the above inequality grows without bound, so no contradiction is obtained. However a significant modification of this proof *could* succeed; we sketch this hopeful proof next.

An important first modification of the above, failed sketch. The idea is to separate the power series for e^n, for each n, $1 \le n \le d$, into a main term, an intermediate term, and a tail, and hope to manipulate the intermediate terms so that a linear combination of them vanishes (and do this in such a way that the tails can be made arbitrarily small):

$$e^n \;=\; \sum_{k=0}^{\infty} \frac{n^k}{k!} \;= \underbrace{\sum_{k=0}^{N} \frac{n^k}{k!}}_{M_N(n)} + \underbrace{\sum_{k=N+1}^{N'} \frac{n^k}{k!}}_{I_{N,N'}(n)} + \underbrace{\sum_{k=N'+1}^{\infty} \frac{n^k}{k!}}_{T_{N'}(n)}.$$

Therefore, assuming $r_0 + r_1 e + r_2 e^2 + \cdots + r_d e^d = 0$ we obtain:

$$r_0 + r_1\big(M_N(1) + I_{N,N'}(1) + T_{N'}(1)\big) + r_2\big(M_N(2) + I_{N,N'}(2) + T_{N'}(2)\big) \\ + \cdots + r_d\big(M_N(d) + I_{N,N'}(d) + T_{N'}(d)\big) = 0,$$

which leads to

$$\underbrace{r_0 + r_1 M_N(1) + r_2 M_N(2) + \cdots + r_d M_N(d)}_{\text{sum of main terms}} \\ = -\Big(\underbrace{r_1 I_{N,N'}(1) + r_2 I_{N,N'}(2) + \cdots + r_d I_{N,N'}(d)}_{\text{sum of intermediate terms}}\Big) \\ - \Big(\underbrace{r_1 T_{N'}(1) + r_2 T_{N'}(2) + \cdots + r_d T_{N'}(d)}_{\text{sum of tails}}\Big)$$

If it were possible to arrange things so that the sum of intermediate terms vanishes, then after multiplying through by $N!$ and letting $M_N^*(n) = \displaystyle\sum_{k=0}^{N} \frac{N!}{k!} n^k$, we would be left with an equation:

$$\big|N! r_0 + r_1 M_N^*(1) + r_2 M_N^*(2) + \cdots + r_d M_N^*(d)\big| \\ = N!\big|r_1 T_{N'}(1) + r_2 T_{N'}(2) + \cdots + r_d T_{N'}(d)\big|,$$

where the expression on the left-hand side is now an integer.

If we look at the leading terms of the tails, as before, and if it were possible to show that the expression $N! r_0 + r_1 M_N^*(1) + r_2 M_N^*(2) + \cdots + r_d M_N^*(d)$ is nonzero, we would have an inequality:

$$0 < |\text{complicated nonzero integer}| \quad < N! \frac{d^{N'+1}}{(N'+1)!} \times e^d \times d \max\{|r_1|, \ldots, |r_d|\}.$$

Then keeping N fixed and letting $N' \to \infty$, the right-hand side approaches 0. Therefore we would obtain:

$$0 < |\text{complicated nonzero integer}| \quad < 1,$$

which is, of course, a contradiction.

The above sketch of a possibly successful proof for the transcendence of e can be made into a formal proof (one which is not as closely aligned with Hermite's original proof as one due to A. Hurwitz in 1893 [16]). This proof is accomplished not by using approximations to the values $e^n, 1 \le n \le d$, obtained by simply dividing the power series representation for e^n into a main term, an intermediate term, and a tail. Rather it requires manipulating the power series for e^z so that the sum of the intermediate terms in the above sketch vanishes at the appropriate values of z. We carry this out in the next chapter.

Exercises

1. a) Derive the following result from Liouville's Theorem: *Let α be a real number. Suppose that for each positive real number c and each positive integer d, there exists a rational number p/q satisfying the inequality*

$$\left|\alpha - \frac{p}{q}\right| < \frac{c}{q^d} \ .$$

Then α is transcendental.

b) Deduce that the number $\ell = \sum_{n=1}^{\infty} 10^{-n!}$ is transcendental.

2. Prove the three versions of the α^β conjecture are equivalent.

3. Hilbert stated part of his seventh problem in a geometric form:
If, in an isosceles triangle, the ratio of the base angle to the angle at the vertex be algebraic but not rational, the ratio between base and side is always transcendental.

a) To which values of the standard trigonometric functions does this conjecture apply?

b) Does this version of Hilbert's conjecture follow from any of the portions of it discussed in this chapter?

4. Derive from one of the results stated in this chapter that if α is a nonzero algebraic number then both $\cos(\alpha)$ and $\sin(\alpha)$ are transcendental.

5. Can Fourier's proof of the irrationality of e be modified to establish the irrationality of e^m, where m is an integer?

6. This exercise outlines a proof of Liouville's Theorem. Fill in the details to justify each step.

a) We only have to consider the case in which α is a real number. (The result is easily seen to be true if α is not real.) Our objective is to demonstrate the existence of a constant c that depends only on α for which the inequality in Liouville's Theorem is satisfied for *all* rational numbers p/q. We may as well restrict ourselves to the case where $\left|\alpha - \frac{p}{q}\right| \leq 1$.

b) Let $P(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$, where $a_d, a_{d-1}, \ldots, a_0 \in \mathbf{Z}$ with $a_d > 0$ be the minimal polynomial for α. Then

$$P\left(\frac{p}{q}\right) = a_d\frac{p^d}{q^d} + a_{d-1}\frac{p^{d-1}}{q^{d-1}} + \cdots + a_1\frac{p}{q} + a_0 \ = \frac{N}{q^d} \ ,$$

where N is a nonzero integer.

c) It follows from b) that

$$\frac{1}{q^d} \leq \left|P\left(\frac{p}{q}\right)\right| \ = \left|P(\alpha) - P\left(\frac{p}{q}\right)\right| \ .$$

d) By the Mean Value Theorem these exists a real number φ between α and p/q such that

$$P(\alpha) - P\left(\frac{p}{q}\right) = P'(\varphi)\left(\alpha - \frac{p}{q}\right) \ .$$

e) Combine the inequalities from c) and d) to conclude the proof. (Remark. Make sure that your constant depends only on α and not on φ.)

Chapter 2

The transcendence of e, π and $e^{\sqrt{2}}$

The fantasy calculation at the end of the last chapter, a fantasy because the linear combination of the intermediate sums, $r_0 + r_r I_{N,N'}(1) + \cdots + r_d I_{N,N'}(d)$, is unlikely to vanish, does give us a goal to pursue: find a series representation for e^z that provides better-than-expected approximations to particular values of e^z. We will see that the hope that it *might* be possible to manipulate the power series for e^z so that when it is divided into a main term, intermediate term, and tail, a linear combination of the intermediate terms vanishes, can be realized. We just need to rethink what we expect from an approximating main term for such a series.

One way to think about the failure of the simple truncation of the power series for e^z to establish the transcendence of e is to realize that we are expecting too much of the series–we are hoping that the truncated series will lead to very good approximations for all values e^n. But we only need good approximations for a few values, rather than *all* values, and we want those approximations to be very good ones. To accomplish this we do not need the intermediate sum to vanish for all values of z but only for those values for which we wish to have good approximations to e^z. This puts us, and Hermite and others, on a new quest: find a polynomial that offers very good approximations to the values under consideration but not particularly good approximations to other values. In particular, we want to find a polynomial that provides a good approximation to e^z at a point $z = a$, but is not necessarily any better than the previous truncation attempt for other values of z. And, in the proof of the transcendence of e we find approximations for each of the powers of e that appears in the assumed nontrivial integral, algebraic equation

$$r_0 + r_1 e + r_2 e^2 + \cdots + r_d e^d = 0.$$

Perhaps surprisingly, this can be accomplished by taking an appropriate integer multiple of the function e^z, which we will see is best thought of as a linear combination of exponential functions. The idea is to take integral combinations of e^z so that the appropriately chosen intermediate term vanishes at each of the values $z = 0, 1, \ldots, d$.

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If we want the intermediate sum to vanish at the values $z = 0, 1, \ldots, d$, the obvious thing to try is to manipulate the power series for e^z so that the intermediate sum is divisible by each of the polynomials $z, z-1, z-2, \ldots, z-d$. Having the intermediate sum divisible by the product $z(z-1)(z-2)\cdots(z-d)$ does not suffice, for reasons we will point out later. The source of the correct integer coefficients is the polynomial:

$$P(z) = z^{p-1}(z-1)^p \cdots (z-d)^p,$$

where the exponent p will be taken to be a sufficiently large prime number in the proof. For now we just point out that $P(z)$ has a zero of order $p-1$ at $z = 0$ and of order p at each of $z = 1, \ldots, d$; the higher order of vanishing at $z = 1, \ldots, d$, together with the requirement that p be a prime number, will play a role in showing that the *small* integer we obtain is nonzero.

If we rewrite the polynomial $P(z)$ as

$$P(z) = c_{p-1}z^{p-1} + c_p z^p + \cdots + c_{(d+1)p-1}z^{(d+1)p-1},$$

and then sum P's 1st through $(p-1)$st derivatives, we obtain the sum:

$$\sum_{n=1}^{p-1} P^{(n)}(z) = \sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=N-p+1}^{N-1} \frac{z^n}{n!} \right) . \tag{2.1}$$

Notice that the right-hand side of this expression equals a sum of terms of the form $N!c_N$ times a portion of the power series for e^z, where the index of the sum, N, runs from $p-1$ to $(d+1)p-1$. This means that we have uncovered a linear combination of the series representation of e^z that has the desired vanishing intermediate sum:

$$\sum_{N=p-1}^{(d+1)p-1} N! c_N e^z = \underbrace{\sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=0}^{N-p} \frac{z^n}{n!} \right)}_{\text{main term } (M_p(z))}$$
$$+ \underbrace{\sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=N-p+1}^{N-1} \frac{z^n}{n!} \right)}_{\text{intermediate term } (I_p(z))} + \underbrace{\sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=N}^{\infty} \frac{z^n}{n!} \right)}_{\text{tail } (T_p(z))} ,$$

provided that we use the convention that an *empty* sum equals 0 (this occurs in the main term when $N = p-1$). As this last point is so important to the proof of the transcendence of e we offer below, we make explicit the main term as:

$$M_p(z) = \sum_{N=p}^{(d+1)p-1} \left(N! c_N \sum_{n=0}^{N-p} \frac{z^n}{n!} \right) .$$

By construction we know that the intermediate term vanishes for $t = 1, 2, \ldots, d$; so for each of these values we have

$$e^t \sum_{N=p-1}^{(d+1)p-1} N! c_N = M_p(t) + T_p(t). \tag{2.2}$$

On the other hand, when $t = 0$ the intermediate term does not vanish, since the polynomial $P(z)$ only has order of vanishing $p - 1$ at $t = 0$. However the tail series clearly vanishes at $t = 0$ so, for $t = 0$, we have the representation:

$$e^0 \sum_{N=p-1}^{(d+1)p-1} N! c_N = M_p(0) + I_p(0) \ . \tag{2.3}$$

The above representations for e^t when $t = 0, 1, \ldots, d$ are the technical tools we need to establish the transcendence of e.

Theorem 2.1. *The number e is transcendental.*

Proof. We begin by again assuming that e is algebraic and so there exist integers $r_0, r_1, \ldots, r_d$, not all zero, such that

$$r_0 + r_1 e + r_2 e^2 + \cdots + r_d e^d = 0 \ .$$

Step 1. When we multiply the equation $r_0 + r_1 e + r_2 e^2 + \cdots + r_d e^d = 0$ by $N! c_N$ and sum from $N = p - 1$ to $(d + 1)p - 1$ we obtain

$$r_0 \sum_{N=p-1}^{(d+1)p-1} N! c_N + r_1 e^1 \sum_{N=p-1}^{(d+1)p-1} N! c_N + \cdots + r_d e^d \sum_{N=p-1}^{(d+1)p-1} N! c_N = 0. \tag{2.4}$$

If we substitute the relationships (2.2) and (2.3) into the equation (2.4) and rearrange terms we obtain the familiar expression:

$$\begin{aligned} r_0\big(M_p(0) + I_p(0)\big) + r_1 M_p(1) + r_2 M_p(2) + \cdots + r_d M_p(d) \\ = -r_1 T_p(1) - r_2 T_p(2) - \cdots - r_d T_p(d) \ , \end{aligned}$$

and therefore

$$\begin{aligned} \big|r_0\big(M_p(0) + I_p(0)\big) + r_1 M_p(1) + r_2 M_p(2) + \cdots + r_d M_p(d)\big| \\ \leq \big|r_1\big|\big|T_p(1)\big| + \big|r_2\big|\big|T_p(2)\big| + \cdots + \big|r_d\big|\big|T_p(d)\big| \ . \end{aligned} \tag{2.5}$$

Step 2. In Step 3 we will show that the expression on the left-hand side of the above equation is a nonzero integer, and, moreover, that it is divisible by the

relatively large integer $(p-1)!$. This is the amazing part of the proof. Before getting there we complete one of the more mundane parts of the proof, we provide an upper bound for the right-hand side of (2.5). Our estimate will, after we complete Step 3, show that (2.5) is an inequality involving a nonzero, positive integer and a number less than 1.

We begin our estimate for the absolute value of the right-hand side of (2.5) by estimating each of the terms $|T_p(t)|$. For $t = 1, 2, \ldots, d$,

$$T_p(t) = \sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=N}^{\infty} \frac{t^n}{n!} \right);$$

the simple change of variables $k = n - N$, and the observation that $\frac{(k+N)!}{k!N!} \geq 1$, yields,

$$N! \sum_{n=N}^{\infty} \frac{t^n}{n!} = \sum_{k=0}^{\infty} \frac{N!}{(k+N)!} t^{k+N} \leq t^N \sum_{k=0}^{\infty} \frac{t^k}{k!} = t^N e^t,$$

It follows from the triangle inequality that

$$|T_p(t)| \leq e^t t^{(d+1)p-1} \sum_{N=p-1}^{(d+1)p-1} |c_N| \ .$$

We next provide an upper bound for the sum $\sum_{N=p-1}^{(d+1)p-1} |c_N|$. To do this we first recall that $z^{p-1}(z-1)^p(z-2)^p \cdots (z-d)^p = \sum_{N=p-1}^{(d+1)p-1} c_N z^N$. So the sum $\sum_{N=p-1}^{(d+1)p-1} |c_N|$ may be bounded by a product of d terms each of which is a bound for the sum of the absolute values of the coefficients of the term $(z-t)^p$, for $t = 1, \ldots, d$. Since $(z-t)^p = \sum_{n=0}^{p} \binom{p}{n} (-t)^{p-n} z^n$, the sum of the absolute values of its coefficients is bounded by

$$\sum_{n=0}^{p} \left| \binom{p}{n} (-t)^{p-n} \right| \leq t^p \sum_{n=0}^{p} \binom{p}{n} = (2t)^p \ .$$

It follows that

$$\sum_{N=p-1}^{(d+1)p-1} |c_N| \leq \prod_{t=1}^{d} (2t)^p \leq \left((2d)^d \right)^p \ .$$

Since $1 \leq t \leq d$ we have

$$|T_p(t)| \leq e^d d^{(d+1)p-1} (2d)^{dp} \ = c_1 \, (c_2)^p \tag{2.6}$$

where the constants c_1 and c_2 are defined by $c_1 = e^d/d$ and $c_2 = d^{2d+1} 2^d$ and depend only on e and its presumed algebraic degree which is at most d.

Thus we have established the following upper bound for the left-hand side of (2.5)

$$\left|r_0\big(M_p(0)+I_p(0)\big)+r_1M_p(1)+r_2M_p(2)+\cdots+r_dM_p(d)\right| \leq c_1\left(\sum_{t=1}^{d}|r_t|\right)(c_2)^p\ . \quad (2.7)$$

Notice that we still have work to do because letting $p\to\infty$ the upper bound on the right-hand side of the above inequality (2.7) is unbounded. It will follow from what we called the amazing part of the proof, which we carry out in the next step, that it is possible to introduce a $(p-1)!$ into the denominator of the right-hand side of (2.7), and still have an integer on the inequality's left-hand side. This will allow us to obtain a contradiction as $p\to\infty$ and therefore conclude that e cannot be algebraic.

Step 3. The integer on the left-hand side of (2.7) is nonzero and is divisible by $(p-1)!$. Specifically we will see that for all sufficiently large prime numbers p,

$$\frac{r_0}{(p-1)!}I_p(0)+\sum_{t=0}^{d}\frac{r_t}{(p-1)!}M_p(t)$$

is a *nonzero* integer.

We establish the above claim in two steps–we first show that the displayed value is an integer, which amounts to showing that $(p-1)!$ divides each term, and we then show that this integer is nonzero, by showing that it is not divisible by p. It will be handy for each of these demonstrations to have the expression for $M_p(t)$ in view so we recall it here:

$$M_p(t)=\sum_{N=p}^{(d+1)p-1}\left(N!c_N\sum_{n=0}^{N-p}\frac{t^n}{n!}\right)=\sum_{N=p}^{(d+1)p-1}\left(c_N\sum_{n=0}^{N-p}\frac{N!}{n!}t^n\right)$$

To establish the first part of the claim we begin by observing that $N\geq p$ and $n\leq(N-p)$ so $\dfrac{N!}{n!}$ is an integer. Moreover, the ratio $\dfrac{N!}{n!(N-n)!}$ is also an integer so $\dfrac{N!}{n!}$ is divisible by $(N-n)!$. Combining this with $N-n\geq p$ yields the stronger than announced result that for each t, $M_p(t)$ is divisible by $p!$. However, from our choice of the polynomial $P(z)$, which led to our intermediate terms, we see that $I_p(0)=(p-1)!c_{p-1}$, which is clearly divisible by $(p-1)!$, thus establishing the first part of the claim.

The second part of the claim follows from the observation that $c_{p-1}\neq 0$, it equals $(-1)^d(d!)^p$, and therefore, if we take $p>d$ it will not divide c_{p-1}. Thus we have

$$\frac{I_p(0)}{(p-1)!}\equiv c_{p-1}\mod p,\ \text{and for each } t\ \ \frac{M_p(t)}{(p-1)!}\equiv 0\mod p,$$

It is now possible to conclude the proof that e is transcendental.

If we divide the inequality (2.7) by $(p-1)!$ we have the inequality

$$0 < \left| \frac{r_0}{(p-1)!} I_p(0) + \sum_{t=0}^{d} \frac{r_t}{(p-1)!} M_p(t) \right| \leq c_1 \left(\sum_{t=1}^{d} |r_t| \right) \frac{(c_2)^p}{(p-1)!},$$

Letting p approach infinity leads to a contradiction, thus establishing the transcendence of e.

As we mentioned in Chapter 1, the above proof is more allied with the one given by Hurwitz in 1893 than with Hermite's original proof.[1] Yet Hermite's proof of 1873 did inspire Lindemann, who just under a decade later established the important theorem:

Theorem 2.2 (The Hermite-Lindemann Theorem [21]). *If α is a nonzero algebraic number then e^{α} is transcendental.*

Sketch of a possible proof of the Hermite-Lindemann Theorem. Suppose α is a nonzero algebraic number and e^{α} is algebraic and satisfies a nontrivial, integral polynomial equation:

$$r_0 + r_1 e^{\alpha} + r_2 e^{2\alpha} + \cdots + r_d e^{d\alpha} = 0.$$

Using the series representation we obtain:

$$\sum_{N=p}^{(d+1)p-1} N! c_N e^{\alpha z} = \underbrace{\sum_{N=p}^{(d+1)p-1} \left(N! c_N \sum_{n=0}^{N-p} \frac{(\alpha z)^n}{n!} \right)}_{\text{main term } (M_p(\alpha z))}$$
$$+ \underbrace{\sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=N-p+1}^{N-1} \frac{(\alpha z)^n}{n!} \right)}_{\text{intermediate term } (I_p(\alpha z))} + \underbrace{\sum_{N=p-1}^{(d+1)p-1} \left(N! c_N \sum_{n=N}^{\infty} \frac{(\alpha z)^n}{n!} \right)}_{\text{tail } (T_p(\alpha z))},$$

where the coefficients c_N are chosen so that the intermediate term, $I_p(\alpha z)$, vanishes at $z = 1, 2, \ldots, d$.

Therefore we have for each $t, 1 \leq t \leq d$,

$$e^{t\alpha} \sum_{N=p}^{(d+1)p-1} N! c_N = M_p(t\alpha) + T_p(t\alpha),$$

while

$$e^{0} \sum_{N=p}^{(d+1)p-1} N! c_N = M_p(0) + I_p(0).$$

[1] See Sections 2.1 and 2.2 of [5] to appreciate the connections between Hermite's proof and those that followed, including Hurwitz's.

This leads to

$$\underbrace{\left|r_0\big(M_p(0)+I_p(0)\big)+r_1M_p(\alpha)+r_2M_p(2\alpha)+\cdots+r_dM_p(d\alpha)\right|}_{\text{term that should lead to a nonzero integer}}$$
$$\leq \underbrace{\left|r_1T_p(\alpha)+r_2T_p(2\alpha)+\cdots+r_dT_p(d\alpha)\right|}_{\text{expression that should be small for } p \text{ large}}.$$

If we could somehow obtain an inequality of roughly the above form, and show the left-hand side is nonzero, we would still need to move from an inequality

$$0 < |\text{nonzero algebraic number }| \leq \text{a small, positive quantity.}$$

to an inequality

$$0 < |\text{nonzero integer }| \leq \text{a small, positive quantity}$$

How to, in general, obtain a rational integer from an algebraic number, is perhaps the only new idea Lindemann had to introduce. We will come back to this outline of the proof.

Algebraic Digression: Obtaining an integer from an algebraic number. Suppose that α is the zero of an irreducible, integral polynomial $P_\alpha(x) = a_dx^d + a_{d-1}x^{d-1} + \cdots + a_0$ with $a_d \neq 0$. If we denote the d zeros of $P(x)$ by $\alpha_1(=\alpha), \alpha_2, \ldots, \alpha_d$ then we have the factorization:

$$P(x) = a_d \prod_{k=1,\ldots,d} \big(x-\alpha_k\big).$$

The algebraic numbers $\alpha_2, \ldots, \alpha_d$ are called the *conjugates* of α and the algebraic norm of α, defined by the product

$$Norm(\alpha) = \prod_{k=1,\ldots,d} \alpha_k$$

is equal to $(-1)^d a_0/a_d$. Thus for any nonzero algebraic number $\alpha, Norm(\alpha)$ is a rational number.

In the particular case that the minimal integral polynomial of α is *monic*, so that its leading coefficient equals 1 (in which case α is said to be an algebraic integer), we see that $Norm(\alpha)$ is a nonzero integer (namely plus-or-minus the constant term of α's minimal, integral polynomial).

It is elementary, and central to transcendence theory, that for any algebraic number α there exists a rational integer δ so that $\delta\alpha$ is an algebraic integer. Any such δ is said to be a denominator for α and if the minimal, integral polynomial for α is the polynomial $P_\alpha(x)$, as above, then $a_d\alpha$ is an algebraic integer. (It is

a zero of the polynomial $Q(x) = x^d + a_{d-1}x^{d-1} + a_d a_{d-2}x^{d-2} + \cdots + a_d^{d-1}a_0$ since $Q(a_d\alpha) = a_d^{d-1}P_\alpha(\alpha) = 0$.)

The proof of the Hermite-Lindemann Theorem (continued)

Assuming e^α is algebraic it is possible to obtain an inequality:

$$0 < |\text{nonzero algebraic number }| \leq \text{a small, positive quantity.}$$

If we then multiply through by the denominator of the algebraic number in the above inequality we get:

$$\begin{aligned} 0 &< |\text{ denominator of the algebraic number} \times \text{nonzero algebraic number }| \\ &\leq |\text{ denominator of the algebraic number}| \times \text{a small, positive quantity.} \end{aligned}$$

Assuming we have a reasonable estimate for the absolute value of the denominator we have just multiplied by we then have:

$$0 < |\text{ nonzero algebraic } \textit{integer}| < \text{ a different, small, positive quantity.}$$

Taking the algebraic norm of the nonzero algebraic integer and estimating the *small, positive quantity* we are led, hopefully, to an inequality:

$$0 < |\text{nonzero integer}| < 1.$$

Such a final contradiction would show that our assumption that e^α is algebraic cannot hold, thus establishing the Hermite-Lindemann Theorem.

The proof of the Lindemann-Weierstrass also uses this approach, where the use of the conjugates of the assumed algebraic values is a bit more elaborate (and subtle). We omit any of these details but refer the interested reader to [3].

Exercises

1. Verify the identity (2.1).

2. Verify the estimate (2.6).

3. a) Show that $\sqrt[3]{2}$ is an algebraic number and find its norm.

b) Find the algebraic norm for each of the zeros of the polynomial $P(X) = 2X^4 + X - 8$. Does your calculation imply that any of these zeros are algebraic integers?

c) Suppose α is an algebraic number whose algebraic norm is a rational integer. Does it follow that α is an algebraic integer?

Chapter 3
Three partial solutions

We recall from Hilbert's address that he considered it to be a very difficult problem to prove that

the expression α^β, for an algebraic base and an irrational algebraic exponent, e. g., the number $2^{\sqrt{2}}$ or e^π, always represents a transcendental or at least an irrational number.

The above quotation is one of the most important comments Hilbert made while posing his seventh problem. We earlier pointed out that Hilbert's seventh problem is usually thought of as being the transcendence of α^β when $\alpha \neq 0, 1$ and the irrational number β are both algebraic. And the numbers that are most often used to illustrate this result are those given above by Hilbert: the numbers $2^{\sqrt{2}}$ and e^π. Yet each of these numbers have additional properties that were exploited to demonstrate their transcendence several years before the solution to the more general problem. In this chapter we discuss in some detail A. O. Gelfond's proof of the transcendence of e^π [9], and then briefly consider R. O. Kuzmin's proof of the transcendence of $2^{\sqrt{2}}$ [17] and then Karl Boehle's generalization of these two results [2].

The first partial solution to Hilbert's seventh problem

Theorem 3.1 (Gelfond, 1929, [9]). *e^π is transcendental.*

Before we look at Gelfond's proof, which is the most technically challenging one we will consider in these notes, we observe that, as Gelfond pointed out at the end of his paper, his method could be modified to establish the following more general result:

Theorem 3.2. *If $\alpha \neq 0, 1$ is algebraic and r is a positive rational number then*

$$\alpha^{\sqrt{-r}} \text{ is transcendental.}$$

This more general result implies the transcendence of both $2^{\sqrt{-2}}$ and $e^\pi = i^{-2i}$ but not the transcendence of the other number Hilbert mentioned, $2^{\sqrt{2}}$.

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Gelfond needed a new idea–the Hermite and Lindemann idea of using a suitable modification of the power series of e^z fails when used to study e^π. And it is easy to see why–any truncation of the power series representation for $e^{\pi z}$ will be a polynomial involving π. According to Lindemann's Theorem π is transcendental so Gelfond could not apply the idea of using this power series to obtain an algebraic number, and so eventually an integer strictly between 0 and 1.

Key new ingredients in Gelfond's proof. At the heart of Gelfond's proof is *not* the power series representation for e^z, or more accurately for the function $e^{\pi z}$, but other polynomial approximations to $e^{\pi z}$. These polynomial approximations are based on the Gaussian integers, so we briefly begin with them.

Ingredient 1. The Gaussian integers are the complex numbers of the form $a+bi$, where a and b are integers. The crucial point about the Gaussian integers is that if e^π is assumed to be an algebraic number then the function $f(z) = e^{\pi z}$ will take on an algebraic value at each of the Gaussian integers. Specifically, if $a + bi$ is a Gaussian integer, then

$$f(a + bi) = e^{\pi(a+bi)} = e^{\pi a} \times e^{i\pi b} = (-1)^b (e^\pi)^a$$

is algebraic. We will discuss the more subtle properties of the Gaussian integers that Gelfond exploited as we present his proof of the transcendence of e^π, but in order to even describe *how* Gelfond used them to give the polynomial approximations to the function $e^{\pi z}$ we need to begin with a way to order them. Gelfond ordered the Gaussian integers by their moduli, and for Gaussian integers with equal moduli by their arguments. This yields the following ordering:

$$z_0 = 0, z_1 = 1, z_2 = i, z_3 = -1, z_4 = -i, z_5 = 1 + i,$$
$$z_6 = -1 + i, z_7 = -1 - i, z_8 = 1 - i, \ldots$$

Ingredient 2. It is possible to approximate the function $e^{\pi z}$ by an infinite series each term of which is a polynomial all of whose zeros are among the (ordered) Gaussian integers (this leads to the so-called *Newton series* of the function $f(z) = e^{\pi z}$). Let $\{z_0 = 0, z_1, z_2, \ldots\}$ denote the (ordered) Gaussian integers and consider the polynomials

$$P_0(z) = 1, \quad P_1(z) = z = z - z_0, \quad P_2(z) = z(z - z_1), \ldots,$$

$$P_k(z) = z(z - z_1) \cdots (z - z_{k-1}).$$

Then we have an *interpolation* to the function $f(z) = e^{\pi z}$ by these polynomials:

$$e^{\pi z} = A_0 P_0(z) + A_1 P_1(z) + A_2 P_2(z) + \ldots + A_n P_n(z) + R_n(z),$$

where the numerical coefficients $A_0, A_1, \ldots A_n$ and the polynomial remainder term $R_n(z)$ have integral representations. Specifically, if γ_n and γ_n' are any

simple, closed curves that enclose the interpolation points $z_0, z_1, \ldots, z_n$ then

$$A_n = \frac{1}{2\pi i} \int_{\gamma_n} \frac{e^{\pi\zeta}}{\zeta(\zeta - z_1)\ldots(\zeta - z_n)} d\zeta$$

and

$$R_n(z) = \frac{P_{n+1}(z)}{2\pi i} \int_{\gamma_n'} \frac{e^{\pi\zeta}}{\zeta(\zeta - z_1)\ldots(\zeta - z_n)(\zeta - z)} d\zeta.$$

We will come back to the important point of choosing these contours in the proof below.

Now that we have this representation of the function $e^{\pi z}$ we can outline Gelfond's proof (we will expand upon and justify each step below).

Outline of Gelfond's Proof

Step 1. Assume e^{π} is algebraic (so the function $e^{\pi z}$ assumes an algebraic value at each Gaussian integer).

Step 2. Show that for n sufficiently large $A_n = 0$. This means that there exists a positive integer N^* so that if $n > N^*, A_n = 0$. This tells us that for all $n > N^*$ we have the representation for the function $e^{\pi z}$:

$$e^{\pi z} = A_0 P_0(z) + A_1 P_1(z) + \cdots + A_{N^*} P_{N^*}(z) + R_n(z).$$

Step 3. It follows upon taking γ_n' to be the circle of radius n, centered at $(0, 0)$, and letting $n \to \infty$, that $R_n(z) \to 0$ for all z. Therefore the function $e^{\pi z}$ may be represented by a polynomial.

Step 4. Conclude that the function e^z is **not** a transcendental function.

This last conclusion contradicts the transcendence of the function e^z and so shows that our initial assumption, that e^{π} is algebraic, cannot hold. Thus e^{π} is transcendental.

Details of Gelfond's Proof

Clearly Step 2, above, is at the heart of Gelfond's proof, so we first focus on this. His demonstration that $A_n = 0$ for all n sufficiently large is ingenious and has two parts. The first part uses analytic tools and it provides an upper bound for $|A_n|$ which depends, in part, on choosing a reasonably short contour of integration γ_n. The second part is entirely algebraic in nature and involves taking the norm of an algebraic number. This is the only part of the proof that uses the assumption that e^{π} is algebraic. Under this assumption each of the expressions A_n is an algebraic number. Using fairly subtle estimates Gelfond finds a denominator for A_n with a manageable absolute value. If $A_n \neq 0$ then multiplying A_n by a denominator and taking the algebraic norm produces an nonzero integer whose absolute value, thanks to the analytic estimate, is less than 1. Thus $A_n = 0$.

Step 2 (Details). Establishing that for n sufficiently large $A_n = 0$.

Part 1. *Analytic Part of Proof–An upper bound for* $|A_n|$. The analytic estimate for $|A_n|$ follows from the representation

$$A_n = \frac{1}{2\pi i}\int_{\gamma_n} \frac{e^{\pi\zeta}}{\zeta(\zeta - z_1)\dots(\zeta - z_n)}d\zeta.$$

Specifically:

$$\begin{aligned}|A_n| &= \left|\frac{1}{2\pi i}\int_{\gamma_n} \frac{e^{\pi\zeta}}{\zeta(\zeta - z_1)\dots(\zeta - z_n)}d\zeta\right| \\ &\le \frac{1}{2\pi} \times (\text{length of the contour } \gamma_n) \times \max_{\zeta\in\gamma_n} \frac{|e^{\pi\zeta}|}{|\zeta||\zeta - z_1|\dots|\zeta - z_n|} \\ &\le \frac{1}{2\pi} \times (\text{length of the contour } \gamma_n) \times \frac{\max_{\zeta\in\gamma_n}|e^{\pi\zeta}|}{\min_{\gamma\in\gamma_n}|\zeta||\zeta - z_1|\dots|\zeta - z_n|}.\end{aligned}$$

To provide a reasonably small upper bound for $|A_n|$ Gelfond needed to understand the possible contours γ_n that would encircle the Gaussian integers appearing in the denominator of the integral representation of A_n. In order to obtain a small upper bound for $|A_n|$ Gelfond needed the length of the contour to be as small as possible, but he also needed each of the terms

$$\max_{\zeta\in\gamma_n}\{|e^{\pi\zeta}|\} \text{ and } \frac{1}{\min_{\zeta\in\gamma_n}|\zeta||\zeta - z_1|\dots|\zeta - z_n|}. \tag{3.1}$$

to be small. Clearly any estimate for either of these quantities will also depend on the choice of the contour of integration.

Since the absolute values of the ordered Gaussian integers are nondecreasing, before specifying γ_n Gelfond needed to estimate $|z_n|$, and so know how large of a contour to use. This estimate is not too difficult to produce. If we let $G(r)$ denote the number of Gaussian integers $x_k + y_k i$ with $x_k^2 + y_k^2 \le r^2$ then it is not too difficult to derive the estimate Gelfond used (one simply shows that $G(r)$ is greater than the area of an appropriately chosen smaller circle and less than the area of an appropriately chosen larger circle (see exercises)). The result is that for $r > \sqrt{2}$,

$$\pi(r - \sqrt{2})^2 \le G(r) \le \pi(r + \sqrt{2})^2.$$

From this it follows, see exercises, that the nth Gaussian integer satisfies:

$$|z_n| = \sqrt{\frac{n}{\pi}} + o(\sqrt{n}), \text{ where } \lim_{n\to\infty}\frac{o(\sqrt{n})}{\sqrt{n}} = 0.$$

The above estimate told Gelfond that for n sufficiently large, he could take the contour of integration to be a circle of radius greater than a constant times $\sqrt{\frac{n}{\pi}}$, Gelfond used the relatively large radius of n. With this contour it is simple to estimate the first expression in (3.1):

$$\max_{\zeta\in\gamma_n}\{|e^{\pi\zeta}|\} \le e^{\pi \max\{Re(\zeta):\zeta\in\gamma_n\}} = e^{\pi n}. \tag{3.2}$$

To estimate the second expression in (3.1) we need an estimate for the minimum distance from each of the first n Gaussian integers, $z_1, z_2, \ldots, z_n$ to the points on the circle γ_n, and we want this minimum to not be too small. A need for such an estimate points to one reason Gelfond took the contour of integration to have a larger radius than would be needed to simply contain the first n Gaussian integers. From the estimate for $|z_n|$, above, we see that for n sufficiently large:

$$|z_n| \le \sqrt{\pi}\sqrt{\frac{n}{\pi}} = \sqrt{n}.$$

So for any $i, 1 \le i \le n$,

$$\min\{|\zeta - z_i| : \zeta \in \gamma_n\} \ge n - \sqrt{n} \ge \frac{1}{2}n, \text{ for } n \text{ sufficiently large.}$$

Therefore we have:

$$\max_{\zeta\in\gamma_n} \frac{1}{|\zeta||\zeta - z_1|\ldots|\zeta - z_n|} = \frac{1}{\min_{\zeta\in\gamma_n}|\zeta||\zeta - z_1|\ldots|\zeta - z_n|} \le \left(\frac{2}{n}\right)^{n+1}$$

Putting all of the above estimates together we obtain, for n sufficiently large,

$$\begin{aligned}
|A_n| &= \left|\frac{1}{2\pi i}\int_{\gamma_n} \frac{e^{\pi\zeta}}{\zeta(\zeta - z_1)\ldots(\zeta - z_n)}d\zeta\right| \\
&\le \frac{1}{2\pi} \times (\text{length of the contour } \gamma_n) \times \max_{\zeta\in\gamma_n} \frac{|e^{\pi\zeta}|}{|\zeta||\zeta - z_1|\ldots|\zeta - z_n|} \\
&\le \frac{1}{2\pi} \times 2\pi n \times e^{\pi n} \times \left(\frac{2}{n}\right)^{n+1} \\
&\le e^{\log n + \pi n - (n+1)\log(n/2)}.
\end{aligned}$$

Warning: If we were to further simplify this estimate to something like $e^{-(1/2)n\log n}$, for n sufficiently large, Gelfond's proof will fail.

Part 2. *Algebraic Part of Proof–A lower bound for $|A_n|$ for those n for which $A_n \neq 0$.* We begin with a simple application of the Residue Theorem that allows us to express A_n as an algebraic number:

$$\begin{aligned}
A_n &= \frac{1}{2\pi i}\int_{\gamma_n} \frac{e^{\pi\zeta}}{\zeta(\zeta - z_1)\ldots(\zeta - z_n)}d\zeta \\
&= \sum_{k=0}^{n} \left(\text{residue of } \frac{e^{\pi z}}{z(z - z_1)\ldots(z - z_n)} \text{ at } z = z_k \right) \\
&= \sum_{k=0}^{n} \frac{e^{\pi z_k}}{\prod_{j=0, j\neq k}^{n} (z_k - z_j)}.
\end{aligned}$$

If for each of the Gaussian integers we use the notation $z_k = x_k + y_k i$, where x_k and y_k are ordinary integers which may be positive, negative, or zero, then

$$A_n = \sum_{k=0}^{n} \frac{e^{\pi z_k}}{\prod\limits_{\substack{j=0 \\ j\neq k}}^{n} (z_k - z_j)} = \sum_{k=0}^{n} \frac{(e^{\pi})^{x_k}(-1)^{y_k}}{\prod\limits_{\substack{j=0 \\ j\neq k}}^{n} (z_k - z_j)}. \tag{3.3}$$

This equation shows that for each n, A_n is an algebraic number because each of the summands in (3.3) is a ratio of algebraic numbers.

Of course the (algebraic) norm of a nonzero algebraic *integer* is a nonzero ordinary integer, but the algebraic norm of an algebraic number that is not an algebraic integer is simply a rational number. In order to obtain an integer from A_n we need to first multiply through by its denominator. The denominator of each of the summands in the above representation of A_n is a product of differences of Gaussian integers. Since $\mathbf{Z}[i]$ is a ring, these denominators are themselves Gaussian integers. We need to better understand both the denominators and numerators in order to find an appropriate integer to multiply A_n by in order to obtain an algebraic integer. It is easier to see what is going on if we simplify our notation. Following Gelfond we put

$$\omega_{n,k} = \prod_{\substack{j=0 \\ j\neq k}}^{n} (z_k - z_j).$$

Then

$$A_n = \frac{(e^{\pi})^{x_0}(-1)^{y_0}}{\omega_{n,0}} + \frac{(e^{\pi})^{x_1}(-1)^{y_1}}{\omega_{n,1}} + \cdots + \frac{(e^{\pi})^{x_n}(-1)^{y_n}}{\omega_{n,n}}. \tag{3.4}$$

A natural thing to try is to let Ω_n equal the product of all of the denominators in (3.4). From the expression

$$|\Omega_n| = \prod_{k=0}^{n} |\omega_{n,k}| = \prod_{k=0}^{n} \prod_{\substack{j=0 \\ j\neq k}}^{n} |(z_k - z_j)|,$$

it is possible to estimate $|\Omega_n|$: it is a product of $n(n-1)$ positive integers each of which is less than $2|z_n|$, and $|z_n| = \sqrt{n/\pi} + o(\sqrt{n})$. But Gelfond employed a more subtle approach.

The ring $\mathbf{Z}[i]$ is a unique factorization domain, so the notion of the least common multiple of a collection of Gaussian integers makes sense. Thus we know that each $\omega_{n,k} = \prod_{j=0, j\neq k}^{n} (z_k - z_j)$ equals a product of irreducible elements from $\mathbf{Z}[i]$, each of which has an absolute value at most $2\sqrt{n}$. If we let $q_1, q_2, \ldots, q_k$ denote the collection of all such irreducible elements of $\mathbf{Z}[i]$ with $|q_j| \leq 2\sqrt{n}$, then

$$|\Omega_n| = \left|L.C.M.\{\omega_{n,0}, \omega_{n,1}, \ldots, \omega_{n,n}\}\right| \leq \left|(q_1 q_1^2 \cdots q_1^{m_1}) \cdots (q_k q_k^2 \cdots q_k^{m_k})\right|,$$

where $m_j = \left[\frac{\log(2\sqrt{n})}{\log |q_j|}\right]$.

In another paper, also published in 1929, Gelfond studied the distribution of the irreducible elements in $\mathbf{Z}[i]$. That result allowed him to conclude that

$$|\Omega_n| \leq e^{\frac{1}{2}n\log n + 163n + O(\sqrt{n})}.$$

We now work with the expression

$$\Omega_n \times A_n = \frac{\Omega_n}{\omega_{n,0}}(e^\pi)^{x_0}(-1)^{y_0} + \frac{\Omega_n}{\omega_{n,1}}(e^\pi)^{x_1}(-1)^{y_1} + \cdots + \frac{\Omega_n}{\omega_{n,n}}(e^\pi)^{x_n}(-1)^{y_n}.$$

to show that $A_n = 0$ for n sufficiently large. It is worth making two observations about this expression:

Observation 1. Each algebraic number $\dfrac{\Omega_n}{\omega_{n,k}}$ is an element of $\mathbf{Z}[i]$, so is an algebraic integer.

Observation 2. In Gelfond's ordering for the elements $z_k = x_k + y_k i$, in $\mathbf{Z}[i]$, x_k may be positive, negative, or zero. Thus each of the numerators in the above expression involves either e^π or $e^{-\pi}$.

These two observations tell us that $\Omega_n \times A_n$ is an integral polynomial expression in $e^\pi, e^{-\pi}$ and i. It is possible to simplify things a bit by multiplying through by an appropriately high power of e^π. Specifically, if we let $r_n = \max_{0\leq k\leq n}\{|x_k|\}$, and note for later use that $r_n \leq \sqrt{n}$, then $(e^\pi)^{r_n}\Omega_n A_n$ is an integral polynomial expression in e^π and i :

$$(e^\pi)^{r_n}\Omega_n A_n = \frac{\Omega_n}{\omega_{n,0}}(e^\pi)^{r_n+x_0}(-1)^{y_0} + \cdots + \frac{\Omega_n}{\omega_{n,n}}(e^\pi)^{r_n+x_n}(-1)^{y_n}.$$

Finally, if we let δ denote a denominator for e^π, we then have the algebraic integer:

$$\begin{aligned} P_n(i,e^\pi) &= (\delta)^{2r_n}(e^\pi)^{r_n}\Omega_n A_n \\ &= (\delta)^{r_n-x_0}\frac{\Omega_n}{\omega_{n,0}}(\delta e^\pi)^{r_n+x_0}(-1)^{y_0} + \cdots + (\delta)^{r_n-x_n}\frac{\Omega_n}{\omega_{n,n}}(\delta e^\pi)^{r_n+x_n}(-1)^{y_n}, \end{aligned}$$

where $P_n(x,y)$ is the obvious integral polynomial. Our goal is to calculate the algebraic norm of the algebraic integer $P_n(i,e^\pi)$, which we will denote by $Norm(P_n(i,e^\pi))$.

Since $P_n(i,e^\pi) \neq 0$ we know that $Norm(P_n(i,e^\pi))$ is a nonzero integer. In order to get a handle on $Norm(P_n(i,e^\pi))$ we denote the conjugates of e^π by $\theta_1(=e^\pi), \theta_2, \ldots, \theta_d$. Then an integral power of $Norm(P_n(i,e^\pi))$ is given by the product:

$$N = P_n(i,\theta_1)P_n(-i,\theta_1)\Big(\prod_{j=2}^{d} P_n(i,\theta_j)\Big)\Big(\prod_{j=2}^{d} P_n(-i,\theta_j)\Big). \tag{3.5}$$

If $A_n \neq 0$ then $N \neq 0$.

We will use our earlier analytic work to provide an upper bound for the first factor, $|P_n(i,\theta_1)|$, and algebraic information about the Gaussian integers to estimate the absolute values of each of the other factors.

Our earlier analytic estimate for $|A_n|$, combined with Gelfond's estimate for $|\Omega_n|$ and the estimates $r_n \le \sqrt{n}$ and $|x_n| \le \sqrt{n}$ yields:

$$|P_n(i,\theta_1)| \le e^{-\frac{1}{2}n\log n+170n}, \text{ provided } n \text{ is sufficiently large.}$$

Each of the other $2d-1$ factors in (3.5) is estimated by the triangle inequality. The most difficult terms to estimate are the ratios $\dfrac{\Omega_n}{\omega_{n,k}}$. We have already seen that in a different paper Gelfond provided an estimate for the numerator $|\Omega_n|$. To provide a lower bound on the denominator, Gelfond used an estimate from another mathematician, Seigo Fukasawa, who, in 1926 [7], showed that

$$|\omega_{n,k}| > e^{\frac{1}{2}n\log n-10n}, \text{ for } n \text{ sufficiently large.}^1$$

From Fukasawa's lower bound we know that, provided n is sufficiently large,

$$\left|\frac{\Omega_n}{\omega_{n,k}}\right| \le e^{\frac{1}{2}n\log n+164n-(\frac{1}{2}n\log n-10n)} \le e^{174n}.$$

Putting all of these estimates together we see that each of the other factors in (3.5) we have:

$$|P_n(\pm i,\theta_j)| \le (n+1)e^{174n}(\delta e^{\pi})^{2r_n} \le e^{175n}, \text{ for } n \text{ sufficiently large.}$$

Finally, we have for the nonzero integer N,

$$0 < |N| \le e^{-\frac{1}{2}n\log n+(2d-1)175n}.$$

The only way to avoid the contradiction presented by the above inequalities is to conclude that our assumption that $A_n \neq 0$ must be wrong. This leads us to the conclusion that for n sufficiently large, say $n > N^*, A_n = 0$. This tells us that for all $n > N^*$ we have the representation for the function $e^{\pi z}$:

$$e^{\pi z} = A_0P_0(z) + A_1P_1(z) + \cdots + A_{N^*}P_{N^*}(z) + R_n(z).$$

[1] Fukasawa's argument has two pieces: To estimate the product of the absolute values of the nonzero Gaussian integers $z_k = x_k + y_k i$, view

$$\log\Big(\prod_{|z_k|\le 2\sqrt{n}} |z_k|\Big) = \sum_{r=\sqrt{x_k^2+y_k^2},} \log r$$

as a Riemann sum. Then estimate this sum by finding *appropriate* disks D_1 and D_2 with:

$$\iint_{D_1} \log r \; dxdy < \sum_{r=\sqrt{x_k^2+y_k^2}} \log r < \iint_{D_2} \log r \; dxdy$$

We take the contour of integration in the integral representation for $R_n(z)$ to be a circle with radius n. Since for any $z, R_n(z) \to 0$ as $n \to \infty$ it follows that $e^{\pi z}$ equals the polynomial $A_0P_0(z) + A_1P_1(z) + \cdots + A_{N^*}P_{N^*}(z)$ and so is not a transcendental function. It follows that e^z is not a transcendental function, which is our long sought contradiction.

Two other partial solutions

The title of this chapter is "Three partial solutions to Hilbert's seventh problem," of which Gelfond's was the first. One year later R. O. Kuzmin gave his partial solution to Hilbert's problem. His paper began:

In December's installment of the journal Comptes Rendus de l'Academie des Sciences de Paris there was an interesting article by A. O. Gelfond, in which the author obtained a new result in the theory of transcendental numbers with the help of extremely clever reasoning. ... The method, which I use here, is closely based on the method of A. O. Gelfond (which I do not know very well, as he published only in Japanese journals, which are inaccessible to me). Perhaps for this reason my methodology is simpler and more elementary. In particular I proceed without complex functional analysis.

Theorem 3.3 (Kuzmin, 1930, [17]). $2^{\sqrt{2}}$ *is transcendental.*

Kuzmin's proof actually established the more general result: For any positive rational number r that is not a perfect square and for any algebraic number $\alpha \neq 0, 1$, $\alpha^{\sqrt{r}}$ is transcendental. For simplicity we only look at his proof of the transcendence of $2^{\sqrt{2}}$.

Both the gross structure, and the nature of the details, in Kuzmin's proof are strikingly similar to Gelfond's proof. In broad outline Kuzmin starts with the assumption that $2^{\sqrt{2}}$ is algebraic and studies this value by considering the real-valued function $2^z = e^{\log(2)z}$. He then approximates 2^z using the Lagrange interpolation formula (not using the Gaussian integers but the numbers $\{a + b\sqrt{2} : a, b$ integers, not both zero$\}$, which he orders as $z_1, z_2, \dots$). Then, using the notation $P_k(z) = (z - z_1)(z - z_2)\dots(z - z_k)$, Kuzmin knew that for each n,

$$2^z = \sum_{k=1}^{n} \frac{P_n(z)}{z - z_k} \frac{2^{z_k}}{P_n'(z_k)} + \frac{P_n(z)}{n!} 2^{\zeta} (\log 2)^n,$$

where ζ lies between the smallest and the largest of z_1 through z_n.

But $P_n'(z) = \sum_{r=1}^{n} \prod_{\ell \neq r} (z - z_\ell)$ which implies that $P_n'(z_k) = \prod_{\ell \neq k} (z_k - z_\ell)$. Therefore, for $z_0 \notin \{z_1, \dots, z_n\}$,

$$2^{z_0} = \sum_{k=1}^{n} \frac{P_n(z_0) 2^{z_k}}{(z_0 - z_k) \prod_{\ell \neq k} (z_k - z_\ell)} + \frac{P_n(z_0)}{n!} 2^{\zeta} (\log 2)^n.$$

Dividing through by $P_n(z_0)$, and rewriting, Kuzmin obtained:

$$\sum_{k=0}^{n} \frac{2^{z_k}}{\prod_{\ell \neq k}(z_k - z_\ell)} = \frac{2^{\zeta}(\log 2)^n}{n!}.$$

Kuzmin takes $z_0 = 0$.

1. Multiplying the left-hand side of this equality by the least common multiple of the denominators and then multiplying by an algebraic denominator yields a nonzero algebraic integer.

2. Taking n to be sufficiently large the right-hand side quantity has a small absolute value.

3. Taking the algebraic norm of the left-hand side produces a nonzero integer whose absolute value is less than 1.

These two estimates contradict each other, so the error term in the Lagrange Interpolation to 2^z equals 0. It follows that the transcendental function$f(z) = 2^z$ is a polynomial function. This contradiction establishes the result.

A couple of years later Karl Boehle published a paper that generalized the results of both Gelfond and Kuzmin, yet still fell far short of *solving* Hilbert's seventh problem. In his paper Boehle acknowledged that his work built on Gelfond and Kuzmin's:

In 1929 A. O. Gelfond demonstrated the transcendence of the number α^β when α is an irrational, neither 0 nor 1, algebraic number and β is a quadratic irrationality. C. L. Siegel showed in a Number Theory Seminar in February 1930 that α^β is transcendental when β is a real quadratic irrationality. R. O. Kuzmin proved this also.

Theorem 3.4 (Boehle, 1933, [2]). *Suppose $\alpha \neq 0, 1$ and β are algebraic numbers, $d = deg(\beta) \geq 2$. Then at least one of the numbers*

$$\alpha^\beta, \ldots, \alpha^{\beta^{d-1}} \text{ is transcendental.}$$

Gelfond's theorem follows from Boehle's upon taking $\beta = \sqrt{-r}$, where r is a positive, rational number. Boehle's theorem then implies that one of the two numbers

$$\alpha^{\sqrt{-r}} \text{ or } \alpha^{(\sqrt{-r})^2} = \alpha^{-r},$$

must be transcendental. Since the second of these numbers is algebraic the first of them, the one Gelfond addressed, must be the transcendental one.

The deduction of Kuzmin's result from Boehle's is similar, with $\sqrt{r}$ replacing $\sqrt{-r}$.

Exercises

1. Show that e^z is a transcendental function.

2. Let a and b be complex numbers. Show that the functions e^{az} and e^{bz} are algebraically independent if and only if a/b is irrational. (We will use this result in the next chapter.)

3. Convince yourself that the estimate for $G(r)$ given above is correct. (Hint: Let $n = G(r)$. Let z_k be one of the first n Gaussian integers and associate with z_k the unit square whose vertices are Gaussian integers and whose lower left corner is the Gaussian integer z_k. We want to take a smaller circle, of radius $r' < r$, so that its area is less than n. It suffices to let $r' = r - \sqrt{2}$. Similarly we want to take a larger radius, $r'' > r$, so than the radius of this larger circle exceeds n. This time it suffices to take $r'' = r + \sqrt{2}$. It follows that

$$\pi(r - \sqrt{2})^2 < G(r) < \pi(r + \sqrt{2})^2.)$$

4. Conclude from the estimate in problem 3 (above) that if z_n denotes the nth Gaussian integer in Gelfond's ordering then

$$|z_n| = \sqrt{\frac{n}{\pi}} + o(\sqrt{n}).$$

(Hint: Suppose $r = |z_n|$ and $m = G(r)$, so $|z_m| = r$ and $n \leq m$. It follows from the above problem that $|z_m| = \sqrt{\dfrac{m}{\pi}} + O(1)$. The result follows because the number of Gaussian integers on the circle of radius of r is at most $4(r + 1)$ so $m - n < 4(r + 1)$.)

5. Verify the equality in line (3.2).

6. Explain why we wrote the N is an *integral power* of the norm of $P_n(i, e^{\pi})$.

7. Convince yourself the number N, in line (3.5), is a nonzero integer.

Chapter 4
Gelfond's solution

Before we consider Gelfond's and Schneider's complete solutions to Hilbert's seventh problem let's look back and see what common elements we can find in Fourier's demonstration of the irrationality of e, the Hermite/Hurwitz demonstration of the transcendence of e, and Gelfond's proof of the transcendence of e^{π}. The first, and most obvious, common feature these proofs share is that they are all proofs by contradiction–the value under consideration is assumed to be rational or algebraic and a contradiction is deduced from that assumption. The second common feature concerns the nature of the contradiction obtained–in each case the deduction leads to a small positive integer, more specifically an integer between 0 and 1. (This was used in Gelfond's proof to show $A_n = 0$ for n sufficiently large. Yet, as seen in the exercises from the previous chapter, Gelfond's proof could be restructured so that the proof's final contradiction is the existence of an integer between 0 and 1.)

In Fourier's proof this integer was produced through a truncation of the power series representation for the number e. In the proof of the transcendence of e this contrarian integer was obtained through simultaneous good rational approximations to $e, e^2, \ldots, e^n$. Gelfond was led to this integer through an examination of the coefficients of a so-called Newton interpolation series to the function $e^{\pi z}$. In each case, the conclusion of the proof relied on establishing two facts about the integer that had been produced: That the integer is nonzero and that its absolute value is less than 1.

Yet these proofs all share another common feature that is only obvious once it is pointed out–each of these proofs are opportunistic in that they rely on a previously known, and explicit, series representation for the number e or the function e^z or the function $e^{\pi z}$. We will see that while both Gelfond and Schneider based their solutions to Hilbert's α^{β} problem on assuming the contrary of what the wished to establish, they did not use previously studied functions. Instead, they each produced a new function that would allow them to exploit the assumed arithmetic nature of the value under consideration to reach the ultimate contradiction. And, although Gelfond and Schneider used it differently, each of the functions they produced depended on an application of the pigeonhole principle, which is more elegantly known as Dirichlet's box principle.

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We begin by stating what Gelfond established (which is the third, equivalent version of the α^β portion of Hilbert's seventh problem we discussed in Chapter 1).

Theorem 4.1 (Gelfond, 1934, [10]). *Suppose that α and β are nonzero algebraic numbers. If*

$$\frac{\log \alpha}{\log \beta}$$

is irrational then it is transcendental.

Before we look at Gelfond's use of the pigeonhole principle to produce an advantageous function let's look at an outline of his proof, which might initially appear to be a bit convoluted. (This sketch, and the complete proof below, are slightly simplified versions of Gelfond's original argument. These were given by Hille [15] in an exposition of Gelfond's argument for an English-speaking audience.)

Detailed Outline of Gelfond's Proof

Step 1. This is Gelfond's point of departure from the earlier opportunistic transcendence proofs. Gelfond used the pigeonhole principle to find integers $c_{k\ell}$, not all zero, so that the function:

$$F(z) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} \alpha^{kz} \beta^{\ell z} = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} e^{\log(\alpha) kz} e^{\log(\beta)\ell z}$$

has the property that $|F^{(t)}(0)|$ is small for a modest range of derivatives.

Step 2. Note that for any t, the derivative $F^{(t)}(z)$ has a particularly simple form:

$$F^{(t)}(z) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} \big(k \log \alpha + \ell \log \beta\big)^t e^{\log(\alpha) kz} e^{\log(\beta)\ell z}.$$

So when $F^{(t)}(z)$ is evaluated at $z = 0$ we obtain an expression:

$$\begin{aligned} F^{(t)}(0) &= \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} (k \log \alpha + \ell \log \beta)^t \\ &= (\log \beta)^t \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} (k \frac{\log \alpha}{\log \beta} + \ell)^t \end{aligned}$$

Thus, using the assumption that $\dfrac{\log \alpha}{\log \beta}$ is algebraic, Gelfond would know that for each t,

$$(\log \beta)^{-t} F^{(t)}(0) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} (k \frac{\log \alpha}{\log \beta} + \ell)^t$$

is an algebraic number.

Each of the values $|F^{(t)}(0)|$ in Step 1, is indeed so small that *if it is nonzero*, then the algebraic norm of the algebraic integer derived from $(\log \beta)^{-t} F^{(t)}(0)$ is a nonzero integer with absolute value less than 1. Thus Gelfond has actually found the original integer coefficients $c_{k\ell}$, not all zero, so that the function:

$$F(z) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} e^{\log(\alpha)kz} e^{\log(\beta)\ell z}$$

has the property that $F^{(t)}(0) = 0$ for a modest range of derivatives.

Step 3. Here is where Gelfond's proof becomes iterative–it only appears to be convoluted until you see its structure. Gelfond used analysis, essentially a clever application of the Maximum Modulus Principle and then the Cauchy Integral Formula, to show that $|F^{(t)}(n)|$ is small for a modest range of integers, n, and for slightly fewer derivatives than in Step 1.

Step 4. Again by taking the algebraic norm of the algebraic integer associated with each of the values

$$(\log \beta)^{-t} F^{(t)}(n) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} (k \frac{\log \alpha}{\log \beta} + \ell) \alpha^{kn} \beta^{\ell n}$$

Gelfond concludes that each of the values $|F^{(t)}(n)|$ is not only small but is equal to 0.

The conclusion of Step 4 implies that the original function has a higher order of vanishing at $z = 0$ than had been discovered in Step 2. This discovery implies that a certain system of equations, with an equal number of equations and unknowns, has a nonzero solution. Since $F(z) \neq 0$, because not all of the coefficients $c_{k\ell}$ are zero and the functions $e^{(\log \alpha)z}$ and $e^{(\log \beta)z}$ are algebraically independent, we see that a certain Vandermonde determinant vanishes. This implies that the ratio $\dfrac{\log \alpha}{\log \beta}$ is rational (contrary to the hypothesis of the theorem).

We look at the details of this proof below.

Finding the advantageous function

Before we give the precise sort of result Gelfond used to find the coefficients $c_{k\ell}$, which yields what we called an advantageous function, let's just look at the general principle behind finding those coefficients. Suppose you have a linear form with real coefficients $a_1, a_2, \ldots, a_k$:

$$L(\mathbf{X}) = a_1 X_1 + a_2 X_2 + \ldots + a_k X_k.$$

Imagine that your goal is to find a nonzero integer vector $\mathbf{X}$, with *small* coordinates, so that $|L(\mathbf{X})|$ is also *small*. This is possible where the two senses of the word *small* are inversely related.

To find the vector $\mathbf{X} = (X_1, \ldots, X_k)$ consider the mapping from $\mathbf{Z}^k$ to $\mathbf{R}$ given by $\mathbf{n} \mapsto L(\mathbf{n})$, so

$$(n_1, n_2, \ldots, n_k) \mapsto a_1 n_1 + a_2 n_2 + \ldots + a_k n_k.$$

Take N to be a positive integer. Then the above mapping maps the set of integer vectors

$$\mathbf{N}(N) = \{(n_1, n_2, \ldots, n_k) : 0 \leq n_i \leq N \text{ for each } i\}$$

into an interval of the real line. The above set contains $(N+1)^k$ vectors and if we divide the interval of the real line containing the image of $\mathbf{N}(N)$ into fewer than $(N+1)^k$ subintervals, then, by the pigeonhole principle, two of the images will be known to lie in the same subinterval. That is, there will exist two vectors $\mathbf{X}_1$ and $\mathbf{X}_2$ in $\mathbf{N}(N)$ so that $L(\mathbf{X}_1)$ and $L(\mathbf{X}_2)$ lie in the same subinterval. If everything is set up correctly we then know that $|L(\mathbf{X}_1 - \mathbf{X}_2)| = |L(\mathbf{X}_1) - L(\mathbf{X}_2)|$ is *small*. Note that the absolute values of the coordinates of the vector $\mathbf{X}_1 - \mathbf{X}_2$ will be at most N, as each of these vectors is an element of $\mathbf{N}(N)$.

We formalize the above discussion as a lemma:

Lemma 4.2. *Let $a_1, a_2, \ldots, a_k$ be real numbers and let $A = \max\{|a_i| : 1 \leq i \leq k\}$. Take any two positive integers N and ℓ so that $(N+1)^k > \ell$. Then there exist rational integers $n_1, \ldots, n_k$ with*

$$0 < \max\{|n_1|, \ldots, |n_k|\} \leq N \tag{4.1}$$

and

$$|a_1 n_1 + a_2 n_2 + \ldots + a_k n_k| \leq \frac{kAN}{\ell}. \tag{4.2}$$

Proof. The lemma follows from the outline above. The only subtlety is to let $-T$ denote the sum of the negative numbers among the a_i and let S denote the sum of the positive numbers among the a_i. Then the mapping $\mathbf{n} \mapsto L(\mathbf{n})$ maps the vectors $\mathbf{N}(N)$ into the real interval $[-kNT, kNS]$, which we subdivide into ℓ intervals of equal length.

Note: There is one trade off between which of the two inequalities (4.1) or (4.2) you wish to have in a simpler form, and another between which of them you wish to be smaller. For example, since the proof of the above lemma requires that $(N+1)^k > \ell$, by way of illustration take $\ell = N^k$. Then (4.2) offers the upper bound:

$$|a_1 n_1 + a_2 n_2 + \ldots + a_k n_k| \leq \frac{kA}{N^{k-1}}.$$

So, as we might expect, the smaller we want the linear form to be the larger we might have to take the integers $n_1, n_2, \ldots, n_k$. Or, put differently, the larger we allow the integers $n_1, \ldots, n_k$ to be the smaller we can make the linear form.

The above simple argument concerning a single linear form with real coefficients can be extended to include the case where the coefficients are complex numbers (each form is viewed as two forms, one involving the real parts of the coefficients and the other the imaginary parts of the coefficients) and to simultaneously include several linear forms (the mapping will then be into $\mathbf{R}^m$, for the appropriate m). Instead of subdividing the image into intervals you subdivide it into $m-$dimensional cubes. The point is to have fewer cubes than image points so two points map into the same cube. This leads to the following result from [15], which we state in a readily applicable form.

Theorem 4.3. *Let $a_{ij}, 1 \leq i \leq n$ and $1 \leq j \leq m$, be complex numbers, with $n > 2m$. Consider the linear forms*

$$L_j(\mathbf{X}) = a_{1j}X_1 + a_{2j}X_2 + \ldots + a_{nj}X_n, \text{ for } 1 \leq j \leq m.$$

Let X be any positive number. Then there exist n rational integers, $N_1, N_2, \ldots, N_n$, not all zero, so that for each j,

$$|a_{1j}N_1 + a_{2j}N_2 + \ldots + a_{nj}N_n| \leq X$$

with

$$\max_{1 \leq i \leq n} \{|N_i|\} \leq [\frac{2^{3/2}nA}{X}]^{\frac{2m}{n-2m}},$$

where $\max\{|a_{ij}|\} \leq A$.

Some Details of Gelfond's Proof

Although Gelfond did not formalize the information about his function that his iterative application of basic analysis and algebra led to, it helps clarify his proof if we codify the result Gelfond obtained in the Steps 1 through 4 (as outlined above) into a single proposition.

Proposition 4.4. *Suppose α and β are nonzero algebraic numbers and that $\dfrac{\log \alpha}{\log \beta}$ is an irrational algebraic number. If K is a sufficiently large positive integer then there exist rational integers $c_{k\ell}, -K \leq k, \ell \leq K$, with $\max\{|c_{k,\ell}|\} \leq 3^{K^2}$, so that the function*

$$F(z) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell}\alpha^{kz}\beta^{\ell z} \tag{4.3}$$

satisfies

$$F^{(t)}(0) = 0 \text{ for } 0 \leq t \leq K^{5/2}. \tag{4.4}$$

Proof. As we qualitatively discussed in our brief look at Steps 1 and 2, above, Gelfond sought to find integers $c_{k\ell}$, not all zero, so that the function:

$$F(z) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} \alpha^{kz} \beta^{\ell z}$$

has the property that the algebraic numbers $|(\log \beta)^{-t} F^{(t)}(0)|$, for $0 \leq t < T$, have algebraic integer equivalents whose algebraic norms are less than 1 in absolute value. We will leave the parameters K and T unspecified until we see what is required of them for this proof to succeed.

In order to find the coefficients $c_{k\ell}$, the expression $(\log \beta)^{-t} F^{(t)}(0)$, for each t, $0 \leq t < T$, is replaced by a linear form. Specifically, we introduce the notation $\mathbf{C}$ for the vector of coefficients $(\ldots, c_{k\ell}, \ldots)$ and consider the linear forms

$$L_t(\mathbf{C}) = (\log \beta)^{-t} F^{(t)}(0) = \sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell} (k \frac{\log \alpha}{\log \beta} + \ell)^t. \tag{4.5}$$

This is a system of T linear forms with complex coefficients $(k \frac{\log \alpha}{\log \beta} + \ell)^t$ and $(2K+1)^2$ unknowns $c_{k\ell}$.

We may apply Theorem 4.3 to find the unknown integers $c_{k\ell}$, provided we have

$$(2K+1)^2 > 2T. \tag{4.6}$$

Before we attempt to specify K or T, let's assume that the above inequality holds and see what is required for us to obtain an appropriate function. We know from Theorem 4.3 that for any $X > 0$ we can find integers $c_{k\ell}$, not all zero, so that

$$\big|L_t(\mathbf{C})\big| < X, \text{ for each } t,$$

where we have the estimate for $C = \max\{|c_{k\ell}|\}$ of,

$$C \leq \left(\frac{2^{3/2}(2K+1)^2 \big(K(\big|\frac{\log \alpha}{\log \beta} + 1\big|)\big)^{T-1}}{X} \right)^{\frac{2T}{(2K+1)^2 - 2T}}. \tag{4.7}$$

This is a rather intimidating inequality, so do not stare at it too long, but simply note that, imagining T and K as having been already chosen, it does offer a relationship between C and X. This relationship is critical at the next step of the proof, where we take an algebraic norm.

If we let δ denote a denominator for the algebraic number $\dfrac{\log \alpha}{\log \beta}$ then $\delta^t L_t(\mathbf{C})$, see (4.5), is an algebraic integer whose norm is easily estimated, at least in terms of our as-yet undetermined entities K, T, X, and C. Let $\eta_1 = \dfrac{\log \alpha}{\log \beta}, \eta_2, \ldots, \eta_d$ denote the conjugates of $\dfrac{\log \alpha}{\log \beta}$ and, temporarily, use notation stressing the dependence of $\delta^t L_t(\mathbf{C})$ on $\dfrac{\log \alpha}{\log \beta}$ by writing, $\delta^t L_t(\mathbf{C}) =$

$P_t\left(\delta\dfrac{\log\alpha}{\log\beta}\right)$, where $P_t(X)$ is the integral polynomial

$$P_t(X) = \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} c_{k\ell}(kX+\delta\ell)^t.$$

Therefore, *if* $P_t\left(\delta\dfrac{\log\alpha}{\log\beta}\right) \neq 0$ the expression

$$N_t = P_t\left(\delta\frac{\log\alpha}{\log\beta}\right)\prod_{j=2,\cdots,d} P_t(\delta\eta_j). \tag{4.8}$$

is a nonzero integer.

The first factor in (4.8) is already known to be relatively small (once we solve for the coefficients $c_{k\ell}$):

$$\left|P_t\left(\delta\frac{\log\alpha}{\log\beta}\right)\right| = \left|\delta^t L_t(\mathbf{C})\right| < \delta^t X.$$

Each of the other factors may be estimated in terms of the other unspecified parameters. To assist us in writing down this estimate we let $c_0 = \max\{|\delta|, |\eta_2|+1, \ldots, |\eta_d|+1\}$. Then,

$$\begin{aligned}\left|P_t(\delta\eta_j)\right| &\le (2K+1)^2 C|\delta|^t (K(|\eta_j|+1))^t\\ &\le 5K^2 C K^t c_0^{2t}\\ &\le 5c_0^{2t}K^{T+2}C\end{aligned}$$

where each of these last two inequalities holds if K is sufficiently large.

Therefore, for each $t,\ 0 \le t < T$,

$$\begin{aligned}\left|N_t\right| &\le |\delta|^T X\left(5c_0^{2T}K^{T+2}C\right)^{d-1}\\ &\le XK^{(d+2)T}C^{d-1}, \quad \text{for } K \text{ sufficiently large.}\end{aligned}$$

If we ignore the appearances of K and T, which will be chosen momentarily, we see that in order to have $\left|N_t\right| < 1$, so that we can conclude that each of the derivatives $F^{(t)}(0) = 0$, we need the product $X \times C^{d-1}$ to be small.

Step 3. To uncover a final bit of information about the relationship between C and the basic parameters of K and T that allows Gelfond's proof to go through, we look at his application of the Maximum Modulus Principle. Gelfond employed a theorem due to Jensen, whose proof relied on the Maximum Modulus Principle, but it is possible to simply appeal to that principle.

Let a be a complex number with $|a| = K^{2/3}$ for which

$$|F(a)| = \max_{|\zeta|=K^{2/3}}\{|F(\zeta)|\}.$$

The function $\dfrac{F(z)}{z^{T-1}}$ is entire, because the parameters will be chosen so that $|N_t| < 1$ so $F(z)$ will have a zero of order $T-1$ at $z=0$. Therefore

$$\left|\frac{F(a)}{a^{T-1}}\right| \leq \max_{|\zeta|=K}\left\{\left|\frac{F(\zeta)}{\zeta^{T-1}}\right|\right\}.$$

Since

$$F(z) = \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} c_{k\ell}\alpha^{kz}\beta^{\ell z}$$

we have the estimate

$$\begin{aligned}\max_{|\zeta|=K}\{\left|F(\zeta)\right|\} &\leq (2K+1)^2 C \max_{|\zeta|=K}\{1,\left|\alpha^{\zeta}\right|\}^K \max_{|\zeta|=K}\{1,\left|\beta^{\zeta}\right|\}^K \\ &\leq (2K+1)^2 C e^{2\max\{1,|\log\alpha|,|\log\beta|\}K^2}.\end{aligned}$$

It follows that

$$\begin{aligned}\left|F(a)\right| &\leq (2K+1)^2 C e^{2\max\{1,|\log\alpha|,|\log\beta|\}K^2}\left(\frac{K^{2/3}}{K}\right)^{T-1} \\ &\leq (2K+1)^2 C e^{2\max\{1,|\log\alpha|,|\log\beta|\}K^2 - \frac{1}{3}(T-1)\log K},\end{aligned}$$

which is a quantity we want to be small.

It is fairly difficult to choose the parameters $K, T, X,$ and C to satisfy not only the requirements we have described so far, but the ones we will encounter below that allow for the completion of the proof. Rather than postpone displaying these parameters until after we have formulated the additional constraints we will impose on them, we display them now in order to simplify our estimates and clarify the remainder of the proof. If we think of K as the free parameter, then T and X are given by:

$$T = [K^2\frac{\log\log K}{\log K}] \text{ and } X = \exp\left(-K^2\frac{\log K}{\log\log K}\right) \tag{4.9}$$

With the above choices it is still a daunting matter to conclude from (4.7) that there are integers $c_{k,\ell}$, not all zero, with $C = \max\{|c_{k,\ell}|\} \leq 3^{K^2}$ so that $F(z)$ satisfies

$$F^{(t)}(0) = 0 \text{ for } 0 \leq t < T. \tag{4.10}$$

By the above application of the Maximum Modulus Principle, we already know that for K sufficiently large

$$\max_{|\zeta|=K^{2/3}}\{|F(\zeta)|\} \leq e^{-\frac{1}{6}K^2\log\log K}.$$

(Note that with the above choices, by allowing the coefficients to be fairly large, we are forcing the absolute values on the linear forms to be very small.)

We now apply the Cauchy Integral Formula to show that $|F^{(t)}(z)|$ is small for a modest range of integers t for all z in a fairly large disc. Consider the integral representation of $F^{(t)}(z_0)$ where we will take $|z_0| \le (1/2)K^{2/3}$,

$$F^{(t)}(z_0) = \frac{t!}{2\pi i}\int_{|\zeta|=K^{2/3}} \frac{F(\zeta)d\zeta}{(\zeta - z_0)^{t+1}}.$$

It follows that for K sufficiently large

$$|F^{(t)}(z)| < e^{-\frac{1}{12}K^2\log\log K} \text{ for } |z| \le \frac{1}{2}K^{2/3},\ 0 \le t \le \frac{K^2}{\log K}.$$

In particular, for any integer $n, -[\frac{1}{2}K^{2/3}] \le n \le [\frac{1}{2}K^{2/3}]$ the algebraic values

$$(\log\beta)^{-t}F^{(t)}(n) = \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} c_{k\ell}(k\frac{\log\alpha}{\log\beta} + \ell)^t\alpha^{kn}\beta^{\ell n} \tag{4.11}$$

satisfy

$$\left|(\log\beta)^{-t}F^{(t)}(n)\right| < e^{-\frac{1}{12}K^2\log\log K} \text{ for } 0 \le t \le \frac{K^2}{\log K}.$$

Step 4. We now use an application of the *algebraic norm idea* to show that each of the algebraic values in (4.11) equals zero. This requires a bit of care because each these expressions involves both positive and negative powers of the algebraic numbers α and β. We first let δ' denote a denominator for all of $\frac{\log\alpha}{\log\beta}, \alpha$ and β. Then for each $n, -[\frac{1}{2}K^{2/3}] \le n \le [\frac{1}{2}K^{2/3}]$ and for each $t, 0 \le t \le \frac{K^2}{\log K}$, we have an algebraic *integer*

$$\begin{aligned}&\left(\delta'\right)^{t+4Kn}(\alpha\beta)^{Kn}(\log\beta)^{-t}F^{(t)}(n)\\ &\qquad = \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} c_{k\ell}(k\delta'\frac{\log\alpha}{\log\beta} + \delta'\ell)^t(\delta'\alpha)^{(K+k)n}(\delta'\beta)^{(K+\ell)n}\end{aligned}$$

If we let

$$Q_{n,t}(X,Y,Z) = \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} c_{k\ell}(kX + \delta'\ell)^tY^{(K+k)n}Z^{(K+\ell)n}, \tag{4.12}$$

then $Q_{n,t}$ is an integral polynomial and

$$Q_{n,t}(\delta'\frac{\log\alpha}{\log\beta}, \delta'\alpha, \delta'\beta) = \left(\delta'\right)^{t+4Kn}(\alpha\beta)^{Kn}(\log\beta)^{-t}F^{(t)}(n).$$

Recall that $\eta_1(=\frac{\log\alpha}{\log\beta}), \eta_2, \ldots, \eta_d$ are the conjugates of $\frac{\log\alpha}{\log\beta}$. We also let $\alpha_1(=\alpha), \alpha_2, \ldots, \alpha_{d_1}$ denote the conjugates of α and $\beta_1(=\beta), \beta_2, \ldots, \beta_{d_2}$ denote

the conjugates of β. Therefore the expression $N_{n,t}$, below, is an integer,

$$N_{n,t} = \left| Q_{n,t}(\delta' \frac{\log \alpha}{\log \beta}, \delta'\alpha, \delta'\beta) \right| \prod_{(j,k,\ell)\neq(1,1,1)} |Q_{n,t}(\delta'\eta_j, \delta'\alpha_k, \delta'\beta_\ell)| \,. \qquad (4.13)$$

We already know that

$$\left|Q_{n,t}(\delta' \frac{\log \alpha}{\log \beta}, \delta'\alpha, \delta'\beta)\right| < e^{-\frac{1}{24}K^2 \log\log K} \text{ for } 0 \le t \le \frac{K^2}{\log K}.$$

Using the explicit representation (4.12) above, and our upper bound for $C = \max\{|c_{k\ell}|\}$, we find that for $(j,k,\ell) \neq (1,1,1)$,

$$\begin{aligned} \left|Q_{n,t}(\delta'\eta_j, \delta'\alpha_k, \delta'\beta_\ell)\right| &\le e^{c_1 t \log K + c_2 K^{5/3}} \\ &\le e^{c_3 K^2}, \qquad \text{provided } t \le \frac{K^2}{\log K}, \end{aligned}$$

where the constants $c_1, c_2, \ldots$, here and below, depend only on α, β, and the choice of logarithms. Therefore

$$\begin{aligned} N_{n,t} &< e^{-\frac{1}{24}K^2 \log\log K} \times \left(e^{c_3 K^2}\right)^{(d-1)d_1 d_2} \\ &< e^{-\frac{1}{30}K^2 \log\log K}, \qquad \text{provided } K \text{ is sufficiently large.} \end{aligned}$$

From this we may conclude that for each n and each t, $N_{n,t} = 0$, which means that for each n and t one of the factors in (4.13) must equal zero. Therefore $Q_{n,t}(\delta' \frac{\log \alpha}{\log \beta}, \delta'\alpha, \delta'\beta) = 0$, from which it follows that $F(z)$ has a zero at each integer $n, -[\frac{1}{2}K^{2/3}] \le n \le [\frac{1}{2}K^{2/3}]$ to order at least $\dfrac{K^2}{\log K}$.

The proof of Proposition 4.4 now follows from another application of the Maximum Modulus Principle followed by an application of the Cauchy Integral Formula. We do not give all of these now-familiar details but only report the outcome.

We begin with a complex number a having $|a| = K^{4/3}$. Then the Maximum Modulus Principle applied to the entire function

$$G(z) = \frac{F(z)}{\displaystyle\prod_{-[\frac{1}{2}K^{2/3}]\le n\le[\frac{1}{2}K^{2/3}]} (n-z)^{\frac{K^2}{\log K}}},$$

on a circle of radius $R = K^{3/2}$, tells us that $|G(a)| \le \max_{|\zeta|=K^{3/2}}\{|G(\zeta)|\}$. Therefore $|F(a)|$ may be bounded by the product:

$$\prod_{-[\frac{1}{2}K^{2/3}]\le n\le[\frac{1}{2}K^{2/3}]} |n-a|^{\frac{K^2}{\log K}} \frac{\max_{|\zeta|=K^{3/2}}\{|F(\zeta)|\}}{\min_{|\zeta|=K^{3/2}}\left\{\displaystyle\prod_{-[\frac{1}{2}K^{2/3}]\le n\le[\frac{1}{2}K^{2/3}]} |n-\zeta|^{\frac{K^2}{\log K}}\right\}}.$$

Each of the terms in the above expression is easily estimated.

1. For n with $-[\frac{1}{2}K^{2/3}] \le n \le [\frac{1}{2}K^{2/3}]$ and a with $|a| = K^{4/3},\ |n-a| < 2K^{4/3}$. Therefore

$$\prod_{-[\frac{1}{2}K^{2/3}]\le n\le[\frac{1}{2}K^{2/3}]} |n-a|^{\frac{K^2}{\log K}} \le \left(2K^{4/3}\right)^{(K^{2/3}+1)\frac{K^2}{\log K}}$$

2. For n with $-[\frac{1}{2}K^{2/3}] \le n \le [\frac{1}{2}K^{2/3}]$ and ζ with $|\zeta| = K^{3/2},\ |n-\zeta| > \frac{1}{2}K^{3/2}$. Therefore

$$\min_{|\zeta|=K^{3/2}} \Big\{ \prod_{-[\frac{1}{2}K^{2/3}]\le n\le[\frac{1}{2}K^{2/3}]} |n-\zeta|^{\frac{K^2}{\log K}} \Big\} \ge (\frac{1}{2}K^{3/2})^{(K^{2/3}+1)\frac{K^2}{\log K}}.$$

3. And finally,

$$\begin{aligned}
\max_{|\zeta|=K^{3/2}}\{|F(\zeta)|\} &\le \max_{|\zeta|=K^{3/2}} \Big\{ \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} |c_{k\ell}||\alpha^{k\zeta}||\beta^{\ell\zeta}| \Big\} \\
&\le (2K+1)^2 \max\{|c_{k\ell}|\} \max_{|\zeta|=K^{3/2}} e^{K|\log\alpha Re(\zeta)|} e^{K|\log\beta Re(\zeta)|} \\
&\le (2K+1)^2 3^{K^2} c_4^{K^{5/2}}
\end{aligned}$$

Putting these estimates together, which we leave to the reader, shows that $|F(a)| \le e^{-\frac{1}{12}K^{8/3}}$, provided we take K to be sufficiently large. Since a was arbitrary we may conclude that

$$|F(z)| < e^{-\frac{1}{12}K^{8/3}}, \quad \text{for all } z, \text{ with } |z| \le K^{4/3}.$$

We use this last estimate in our application of the Cauchy Integral Formula. Specifically we know that:

$$F^{(t)}(0) = \frac{t!}{2\pi i}\int_{|\zeta|=K^{4/3}} \frac{F(\zeta)}{\zeta^{t+1}} d\zeta.$$

A fairly routine calculation yields:

$$|F^{(t)}(0)| < e^{-\frac{1}{24}K^{8/3}}, \quad \text{for a large range of } t, \text{ for example for } 0 \le t \le K^{5/2}.$$

The proof is completed by representing

$$(\delta)^t(\log\beta)^{-t}F^{(t)}(0) = \sum_{k=-K}^{K}\sum_{\ell=-K}^{K} c_{k\ell}(k\delta\frac{\log\alpha}{\log\beta} + \delta\ell)^t,$$

as an integral polynomial expression

$$P_t\left(\delta\frac{\log\alpha}{\log\beta}\right),$$

and showing that the integer N_t defined by

$$N_t = \left| P_t\left(\delta \frac{\log \alpha}{\log \beta}\right)\right| \left| \prod_{j\neq 1} P_t\left(\delta\eta_j\right)\right|$$

equals zero. This last conclusion does, indeed, establish the proposition.

The conclusion of Gelfond's solution

The above proposition tells us that $F^{(t)}(0) = 0$ for $0 \leq t \leq K^{5/2}$. We translate this conclusion into a system of equations

$$\sum_{k=-K}^{K} \sum_{\ell=-K}^{K} c_{k\ell}(k\frac{\log \alpha}{\log \beta} + \ell)^t = 0,\ 0 \leq t \leq K^{5/2}$$

Since this system of equations has a nonzero solution, namely the $(2K+1)^2$ coefficients $c_{k\ell}$, we know that any $(2K+1)^2-$rowed determinant of the matrix associated with the above system of equations must vanish. In particular,

$$\det \left| \left(k\frac{\log \alpha}{\log \beta} + \ell\right)^t \right| = 0,\ \ -K \leq k, \ell \leq K,\ 0 \leq t \leq 4K(K+1) = (2K+1)^2 - 1.$$

The above determinant is a Vandermonde determinant, so it vanishes if and only if two of its columns are equal. This is the same as the condition that for two pairs of integers $(k,\ell) \neq (k',\ell')$, $k\dfrac{\log \alpha}{\log \beta} + \ell = k'\dfrac{\log \alpha}{\log \beta} + \ell'$, which implies that $\dfrac{\log \alpha}{\log \beta} = \dfrac{\ell - \ell'}{k' - k}$ is a rational number, contrary to Hilbert's, and Gelfond's hypothesis.

Exercises

1. Let $a_1, a_2, \ldots, a_n$ be complex numbers.

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & \ldots & 1 \\ a_1 & a_2 & \ldots & a_n \\ a_1^2 & a_2^2 & \ldots & a_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \ldots & a_n^{n-1} \end{pmatrix}.$$

Show that the determinant of the above matrix, a so-called Vandermonde determinant, equals 0 if and only if $a_i = a_j$ for some $i \neq j$.

2. Verify that with the choices of parameters (4.9) the inequality (4.7) allows us to assume that $C = 3^{K^2}$, provided K is sufficiently large.

3. Did Gelfond use his assumption that $\frac{\log \alpha}{\log \beta}$ is algebraic in Step 1 of his proof. If so, how?

Chapter 5
Schneider's solution

In this chapter we will briefly examine Schneider's solution [25] to Hilbert's seventh problem, which appeared within a few months of Gelfond's. (The story goes that Schneider learned of Gelfond's solution as he was submitting his own paper for publication.) Like Gelfond's proof, Schneider's depended on an application of the pigeonhole principle, elementary complex analysis, and the fundamental fact that the algebraic norm of a nonzero algebraic integer is a nonzero rational integer. However, Schneider did not apply the pigeonhole principle to solve a system of *inequalities*, and then show that these inequalities implied the vanishing of a function at certain points (with multiplicities) as Gelfond had. Rather, he directly solved a system of *equalities*, indeed a system of homogeneous linear equations. This idea has been attributed to Schneider's thesis advisor, C. L. Siegel (1929) [27], and can be traced back to Axel Thue (1909) [30]. This approach allowed Schneider to find an entire function with prescribed zeros, without having to iterate the use of an analytic estimate and of algebraic norms.

Before we explain the consequence of the pigeonhole principle Schneider used, we state a proposition which follows from that application. We will see that the deduction of this proposition is significantly more straightforward than the deduction of the analogous proposition in Gelfond's solution, even if its statement is not. And it is easy to understand why this is necessarily the case.

At the center of Schneider's proof is a function $F(z) = P(z, e^{z \log \alpha})$, where P is an integral polynomial. Schneider used the pigeonhole principle to find the coefficients of P so that for a range of positive integers a and b,

$$0 = F(a + b\beta) = P(a + b\beta, e^{(a+b\beta) \log \alpha}).$$

Finding a polynomial P so that the function $P(z, e^{z \log \alpha})$ has the above zeros will depend on the assumption that α^β is algebraic. And, since $P(a + b\beta, e^{(a+b\beta) \log \alpha})$ is an expression involving all of α, β, and α^β, the coefficients we find must necessarily depend on these numbers. This dependence is reflected in the statement of the proposition.

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Proposition 5.1. *Suppose α and β are algebraic numbers with $\alpha \neq 0, 1$ and β irrational. Further assume that α^β is algebraic. Let $d = [\mathbf{Q}(\alpha, \beta, \alpha^\beta) : \mathbf{Q}]$ and take m be a positive integer. Put $D_1 = [\sqrt{2d}m^{3/2}]$ and $D_2 = [\sqrt{2d}m^{1/2}]$. Then if m is sufficiently large there exist rational integers $c_{k\ell}, 0 \leq k \leq D_1 - 1,\ 0 \leq \ell \leq D_2 - 1$, not all zero, such that the function*

$$F(z) = \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} z^k \alpha^{\ell z} \tag{5.1}$$

satisfies

$$F(a + b\beta) = 0 \text{ for } 1 \leq a, b \leq m. \tag{5.2}$$

Moreover, there exists a constant c_0 depending on $\alpha, \beta, \alpha^\beta$, and our choice of an algebraic integer θ with $\mathbf{Q}(\alpha, \beta, \alpha^\beta) = \mathbf{Q}(\theta)$, so that the integers $c_{k\ell}$ satisfy

$$0 < \max |c_{k\ell}| \leq c_0^{m^{2/3} \log m}. \tag{5.3}$$

Before we prove this proposition we note how it differs in two significant ways from Proposition 4.4. First, the function vanishes at several points, and, second, there is no reference to the derivatives of the function. Because of this later observation, Schneider's method can be applied to some problems that are not immediately approachable by Gelfond's method. The proof of this proposition depends on an elementary result to guarantee the existence of the unknown coefficients $c_{k\ell}$.

Suppose we wish to find a nonzero integral solution to a homogeneous system of M linear equations in N unknowns:

$$\begin{aligned} a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N &= 0 \\ a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N &= 0 \\ &\vdots \\ a_{M1}X_1 + a_{M2}X_2 + \cdots + a_{MN}X_N &= 0 \end{aligned}$$

where the coefficients a_{mn} are *integers*, not all equal to 0.

The matrix of coefficients (a_{mn}) may be viewed as a mapping from $\mathbf{R}^N$ to $\mathbf{R}^M$ so basic linear algebra tells us that if $N > M$ then there is a nonzero vector in the mapping's kernel, thus there are *real* solutions to the above system of equations. But the result we seek is that if $N > M$ there are *integral* solutions to this system of equations whose absolute values may be bounded from above. We will see that this bound will depend only on M, N, and the absolute values of the coefficients a_{mn}. It is perhaps surprising that the deduction of this result is no more difficult than the deduction of result Gelfond employed, Theorem 4.3.

We start with the notation $A = \max\{|a_{mn}| : 1 \leq m \leq M,\ 1 \leq n \leq N\}$. We want to use the system of equations to map integral vectors in $\mathbf{Z}^N$ into $\mathbf{Z}^M$, so we consider the $M \times N$ matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{22} & \dots & a_{2N} \\ \vdots & \vdots & \vdots & \vdots \\ a_{M1} & a_{M2} & \dots & a_{MN} \end{pmatrix}.$$

Then we are searching for a nonzero vector

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_N \end{pmatrix} \in \mathbf{Z}^N$$

satisfying $\mathbf{AX} = \mathbf{0}$, or equivalently, $\mathbf{X}$ is a nonzero solution to the system of equations above.

Suppose we take a cube of vectors $\mathcal{D}$ in $\mathbf{Z}^N$ and using the matrix $\mathbf{A}$, map the vectors in $\mathcal{D}$ into a rectangular box of vectors $\mathcal{R}$ in $\mathbf{Z}^M$. If there are fewer integral vectors in the range set $\mathcal{R}$ than in the domain set $\mathcal{D}$, then there must exist two *distinct* integer vectors $\mathbf{x_1}$ and $\mathbf{x_2}$ in $\mathcal{D}$ that get mapped to the *same* vector in $\mathcal{R}$. That is, $\mathbf{Ax_1} = \mathbf{Ax_2}$. Thus we see that $\mathbf{X} = \mathbf{x_1} - \mathbf{x_2}$ is a *nonzero* integer solution to $\mathbf{AX} = \mathbf{0}$. Moreover, since the vectors $\mathbf{x_1}$ and $\mathbf{x_2}$ are both from the domain cube $\mathcal{D}$, we can bound the size of the largest component of the solution vector $\mathbf{x_1} - \mathbf{x_2}$.

To carry this out we let $X \geq 1$ be an integer and define the N-dimensional domain cube $\mathcal{D}(X)$ by

$$\mathcal{D}(X) = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} \in \mathbf{Z}^N : 0 \leq x_n \leq X \text{ , for all } n = 1, 2, \dots, N \right\}.$$

$\mathcal{D}(X)$ contains $(1 + X)^N$ vectors.

The matrix $\mathbf{A}$ maps $\mathcal{D}(X)$ into an easily described subset of $\mathbf{Z}^M$. The description of this set is simplified if for any integer k we put $k^+ = \max\{0, k\}$ and $k^- = \max\{0, -k\}$. We can then define the appropriate set by

$$\mathcal{R}(X) = \left\{ \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{pmatrix} \in \mathbf{Z}^M : -X \sum_{n=1}^{N} a_{mn}^- \leq y_m \leq X \sum_{n=1}^{N} a_{mn}^+, \ 1 \leq m \leq M \right\}.$$

It is easy to verify that $\mathbf{A}(\mathcal{D}(X)) \subseteq \mathcal{R}(X)$. A calculation shows that the cardinality of $\mathcal{R}(X)$ is at most $(1 + XAN)^M$, where we recall that $A = \max\{|a_{mn}|\}$.

By the pigeonhole principle, if there are more integral vectors in $\mathcal{D}(X)$ than there are integral vectors in $\mathcal{R}(X)$ then $\mathbf{A}$ must map two vectors to the same vector. Explicitly, if

$$(1+X)^N > (1+XAN)^M, \tag{5.4}$$

then $\mathbf{A}$ will map two distinct vectors $\mathbf{x_1}, \mathbf{x_2} \in \mathcal{D}(X)$ to the same vector in $\mathcal{R}(X)$. Thus we have that $\mathbf{A}(\mathbf{x_1} - \mathbf{x_2}) = \mathbf{0}$, where $\mathbf{x_1} - \mathbf{x_2}$ is a *nonzero* integer vector. Moreover, each coordinate of both $\mathbf{x_1}$ and $\mathbf{x_2}$ is an element of the set $\{0, 1, \ldots, X\}$, so the maximum absolute value of the difference of any two of their coordinates must be less than or equal to X.

We are naturally led to the following observation: Given that condition (5.4) must hold for us to apply the pigeonhole principle we next seek the smallest possible X that satisfies that condition as this will lead to a good estimate for the size of the solutions to our original system of equations. It can be shown, and it is an exercise below to do so, that given positive integers A, M, and N, with $N > M$, the value

$$X = \left[(AN)^{\frac{M}{N-M}} \right], \tag{5.5}$$

suffices.

The above discussion establishes the following theorem that we apply to establish Proposition 5.1

Theorem 5.2 (Siegel's Lemma). *Let $\mathbf{A} = (a_{mn})$ be a nonzero $M \times N$ matrix having integer entries and let $A = \max\{|a_{mn}| : 1 \le m \le M,\ 1 \le n \le N\}$. Assume $A \ge 1$. If $N > M > 0$ then there exists a nonzero vector*

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{pmatrix} \in \mathbf{Z}^N,$$

satisfying $\mathbf{AX} = \mathbf{0}$ with

$$\max\{|X_1|, \ldots, |X_N|\} \le (AN)^{\frac{M}{N-M}}. \tag{5.6}$$

Note: Siegel's Lemma enabled Schneider to describe the function he desired for his proof. This theorem does not explicitly yield the integral solutions $X_1, \ldots, X_N$; it only establishes that they exist and that their absolute values may be estimated. However, Schneider would know that for any pair of integers a and b

$$\begin{aligned} F(a+b\beta) &= \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} (a+b\beta)^k e^{\ell \log \alpha (a+b\beta)} \\ &= \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} (a+b\beta)^k \alpha^{a\ell} \alpha^{\beta b \ell}, \end{aligned}$$

is an integral polynomial expression involving $\alpha, \beta,$ and α^β and so is an algebraic number.

There is one small twist to applying Siegel's Lemma to obtain the appropriate function–the coefficients in Schneider's system of equations are not rational integers, as is required in order to obtain integral solutions, but algebraic numbers. This does not present too great of an obstacle as we simply represent each algebraic number as a linear combination of powers of a primitive element for a field containing all of the algebraic numbers under consideration. We then set each coefficient in this linear expression equal to zero.

Outline of the proof of Proposition 5.1

Step 1. Translate the condition that $F(a + b\beta) = 0$ for $1 \le a, b \le m$ into a system of m^2 linear equations with *algebraic coefficients*.

Step 2. Using a primitive element for the number field $K = \mathbf{Q}[\alpha, \beta, \alpha^\beta]$, translate the condition $F(a + b\beta) = 0$ for $1 \le a, b \le m$ into a larger system of equations with rational integral coefficients.

Step 3. Apply Siegel's Lemma to obtain the appropriate function $F(z)$.

Details of the proof.

Step 1. We begin by representing our desired function $F(z)$ with undetermined coefficients $c_{k\ell}$ and unspecified degrees D_1 and D_2 :

$$F(z) = \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} z^k e^{(\log \alpha)\ell z}.$$

In order to translate the vanishing of $F(z)$ at all of the desired points $a + b\beta$ into a rather explicit homogeneous system of linear equations with integer coefficients it helps to introduce informative notation for the coefficients of the system of linear equations corresponding to the conditions

$$F(a + b\beta) = 0 \quad \text{for} \quad 1 \le a \le m, 1 \le b \le m \ .$$

Since

$$\begin{aligned} F(a + b\beta) &= \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} (a + b\beta)^k e^{(\ell \log \alpha)(a+b\beta)} \\ &= \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} \underbrace{(a + b\beta)^k (\alpha)^{\ell a} (\alpha^\beta)^{\ell b}}_{\text{coefficients}} \underbrace{c_{k\ell}}_{\text{unknowns}} = 0 \ , \end{aligned}$$

we see that for each choice of integers k, ℓ, a, b we need to understand the algebraic number

$$(a + b\beta)^k e^{(\log \alpha)\ell a} e^{(\beta \log \alpha)\ell b} = (a + b\beta)^k (\alpha)^{\ell a} (\alpha^\beta)^{\ell b}. \tag{5.7}$$

Step 2. Let θ be a primitive element for the field $K = \mathbf{Q}(\alpha, \beta, \alpha^\beta)$, which is also an algebraic integer. Then there are rational polynomials p_α, p_β and p_{α^β} of degrees at most $d-1$ so that

$$\alpha = p_\alpha(\theta), \beta = p_\beta(\theta) \text{ and } \alpha^\beta = p_{\alpha^\beta}(\theta).$$

Thus a typical term in the summand representing $F(a+b\beta)$ may be rewritten as:

$$(a+b\beta)^k e^{(\log\alpha)\ell a} e^{(\beta\log\alpha)\ell b} = (a + bp_\beta(\theta))^k (p_\alpha(\theta))^{\ell a} (p_{\alpha^\beta}(\theta))^{\ell b}.$$

We let δ denote the least common multiple of the denominators of the coefficients of p_α, p_β and p_{α^β}. Then

$$\delta^{D_1+2D_2m} F(a+b\beta)$$

may be rewritten to involve only rational integers and the algebraic integer θ. And although the above expression will involve powers of θ greater than $d-1$, using the following lemma each of these may be rewritten as a linear combination of $1, \theta, \ldots, \theta^{d-1}$ with coefficients of predictable absolute values. The statement of this lemma requires that we introduce a new concept, the height of an algebraic number. For an arbitrary algebraic number α of degree $\deg(\alpha) = d$, let

$$P(x) = c_d x^d + c_{d-1} x^{d-1} + \cdots + c_0 \in \mathbf{Z}[x],$$

denote the minimal polynomial for α; thus $\gcd(c_0, c_1, \ldots, c_d) = 1$. We define the *height of* α, denoted by $H(\alpha)$, to be the height of its minimal polynomial. That is,

$$H(\alpha) = H(P) = \max\{|c_0|, |c_1|, \ldots, |c_d|\} .$$

Lemma 5.3. *Suppose $\beta_1, \beta_2, \ldots, \beta_L$ are elements of $\mathbf{Q}(\theta)$, where θ is an algebraic integer of degree d and of height $H(\theta)$. If for each $l = 1, 2, \ldots, L$,*

$$\beta_l = r_{l1} + r_{l2}\theta + \cdots + r_{ld}\theta^{d-1} ,$$

where each r_{lj} is a rational number satisfying $|r_{lj}| \le B_l$ for some bound B_l, then

$$\beta_1\beta_2\cdots\beta_L = r_1 + r_2\theta + \cdots + r_d\theta^{d-1} ,$$

with rational coefficients r_j satisfying

$$\max_{1\le j\le d}\{|r_j|\} \le d^L B_1 B_2 \cdots B_L \Big(2H(\theta)\Big)^{dL}$$

Moreover, if $\mathrm{den}(\beta_l)$ denotes the least common multiple of the denominators of the rational coefficients $r_{l1}, r_{l2}, \ldots, r_{ld}$, then each rational number r_j has a denominator of the form

$$\mathrm{den}(\beta_1)\mathrm{den}(\beta_2)\cdots\mathrm{den}(\beta_L) .$$

This lemma tells us that a typical summand in $\delta^{D_1+2D_2m}F(a+b\beta)$ may be rewritten as,

$$\delta^{(D_1-k)+(D_2m-\ell a)+(D_2m-\ell b)}(\delta(a+bp_\beta(\theta)))^k(\delta p_\alpha(\theta))^{\ell a}(\delta p_{\alpha^\beta}(\theta))^{\ell b} \\ = a_1(k,\ell,a,b)+a_2(k,\ell,a,b)\theta+\cdots+a_d(k,\ell,a,b)\theta^{d-1} \quad (5.8)$$

where the integers $a_1, a_2, \ldots, a_d$ satisfy

$$\max_{1\le r\le d}|a_r(k,\ell,a,b)| \le c_1^{D_1\log(\max\{a,b\})+D_2\max\{a,b\}}, \quad (5.9)$$

c_1, and the other constants $c_2, \ldots$ below, depend only on α, β, and our choice of θ, but not on any of the parameters. (For an explicit value for c_1 see the exercises.)

Thus pulling all of our observations together, we see that for each pair of integers a and b, we have

$$\delta^{D_1+2D_2m}F(a+b\beta) = A_1+A_2\theta+\cdots+A_d\theta^{d-1}\ ,$$

where each integer $A_j = A_j(a,b)$ can be expressed as

$$A_j(a,b) = \sum_{k=0}^{D_1-1}\sum_{\ell=0}^{D_2-1} a_j(k,\ell,a,b)c_{k\ell}\ .$$

Since the numbers $1, \theta, \theta^2, \ldots, \theta^{d-1}$ are $\mathbf{Q}$-linearly independent, it follows that $\delta^{D_1+2D_2m}F(a+b\beta) = 0$, so $F(a+b\beta)$ equals 0, if and only if each of the associated quantities $A_1, A_2, \ldots, A_d$ equals 0. Therefore we can replace each single linear equation $F(a+b\beta) = 0$ involving algebraic coefficients with d linear equations involving only integer coefficients. Namely,

$$A_1(a,b) = 0\ ,\quad A_2(a,b) = 0\ ,\ldots,\ A_d(a,b) = 0\ .$$

Step 3. For each pair a, b, if we set each of the associated linear forms $A_1, A_2, \ldots, A_d$ equal to 0, we obtain a homogeneous system of dm^2 linear equations in D_1D_2 unknowns. By Siegel's Lemma, if

$$D_1D_2 > dm^2\ ,$$

then there exist integers $c_{k\ell}$, not all zero, that are a solution to the linear system

$$A_j(a,b) = \sum_{k=0}^{D_1-1}\sum_{\ell=0}^{D_2-1} a_j(k,\ell,a,b)c_{k\ell} = 0\ ,$$

for $j = 1, 2, \ldots, d$, $a = 1, \ldots, m$, and $b = 1, \ldots, m$, such that for each k and ℓ,

$$|c_{k\ell}| < \left(D_1D_2c_1^{D_1\log m+D_2m}\right)^{\frac{dm^2}{D_1D_2-dm^2}}\ .$$

In order to simplify this upper bound we fix a relationship between the parameters D_1, D_2, and m. The natural thing to try is to take $D_1 \log m$ equal to $D_2 m$, as these two expressions contribute equally to the estimate for the absolute values of the coefficients $c_{k\ell}$. The inclusion of a logarithmic term is necessary in many transcendence proofs, for example in Gelfond's solution to Hilbert's seventh problem, but in Schneider's somewhat less delicate proof we can ignore the relatively slow-growing $\log m$ factor. We choose D_1 and D_2 such that

$$D_1 D_2 = 2dm^2 \quad \text{and} \quad D_1 = D_2 m \ .$$

If we imagine that m is our free parameter, we solve for D_1 and D_2 and obtain:

$$D_1 = \sqrt{2d}\, m^{3/2} \quad \text{and} \quad D_2 = \sqrt{2d}\, m^{1/2} \ ,$$

with the additional understanding that we will henceforth take m such that these quantities are integers (i.e., take m always of the form $m = 2dn^2$ where n is an integer).

We note that indeed $D_1 D_2 = 2dm^2 > dm^2$ as required by Siegel's Lemma. So applying that lemma we see that for m sufficiently large there exist integers $c_{k\ell}$, not all zero, satisfying

$$|c_{k\ell}| < c_0^{m^{3/2} \log m} \ , \tag{5.10}$$

so that if $P(x,y) = \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} x^k y^\ell$, and $F(z) = P(z, e^{\log \alpha z})$, then $F(z)$ is a nonzero function with the property that for each $a = 1, \ldots, m$ and $b = 1, \ldots, m$,

$$F(a + b\beta) = 0 \ .$$

This completes the proof of Proposition 5.1.

Once we have been assured that a function with our prescribed zeros exists, we need a nonzero value of the function that leads to a nonzero algebraic integer whose norm is less than 1. This requires three things: a nonzero algebraic number derived from a value of the function, an upper bound for the absolute value of this algebraic number, and information about the algebraic number's conjugates. An important observation that will assist us in meeting all of these requirements is that since β is irrational, we see that $a + b\beta = a' + b'\beta$ if and only if $a = a'$ and $b = b'$. Therefore, by our construction, $F(z)$ has at least m^2 *distinct* zeros, namely, at $z = a + b\beta$, for $1 \le a \le m$ and $1 \le b \le m$.

The conclusion of the proof

Before we discuss *how* to find a nonzero value for the function $F(z)$ let's examine how such a nonzero value leads to a completion of the proof. We will return to guarantee the existence of an appropriate nonzero value in the next section.

Using algebraic conjugates. Using a fairly complicated determinant argument, see below, Schneider proved that there exists a pair of integers a^*, b^*, with

$1 \le a^*, b^* < 4m$, so that $F(a^* + b^*\beta) \neq 0$. In this section we use the algebraic norm to obtain a nonzero integer from the nonzero algebraic number $F(a^* + b^*\beta)$ whose absolute value is less than 1. Although we have already seen such an argument in some detail we will give fairly complete details here. However we will not explicitly display the dependence on $\alpha, \beta,$ or α^β, and consequently on θ and d, in our estimates. Rather we continue to absorb these explicit dependencies into consecutively numbered constants $c_2, c_3, \ldots$.

We begin by letting $m^* = \min\{a^*, b^*\}$, with the property that

$$F(a + b\beta) = 0 \text{ for } 1 \le a, b < m^* \text{ and } F(a^* + b^*\beta) \neq 0;$$

and we use what Schneider showed, that $m \le m^* < 4m$ (see next section).

Recalling our earlier notation if we recompute the estimates required to apply Siegel's Lemma using the specific numbers a^* and b^*, which give rise to a nonzero value for the function $F(z)$, instead of with general a and b satisfying $1 \le a, b \le m$ we have

$$\delta^{D_1+2D_2m^*} F(a^* + b^*\beta) = A_1^* + A_2^*\theta + \cdots + A_d^*\theta^{d-1}, \tag{5.11}$$

where the integers $A_j^* = A_j^*(a^*, b^*)$ satisfy

$$\begin{aligned}\max_{1\le j\le d}\{|A_j^*|\} &\le \delta^{D_1+2D_2m^*} \max\{|c_{k\ell}|\} \max\{|a_j(k,\ell,a^*,b^*)|\} \\ &\le c_2^{D_1+D_2m^*} c_0^{m^{3/2}\log m} c_1^{D_1 \log m^* + D_2 m^*} \\ &\le c_3^{m^{3/2}\log m}, \text{ since } m^* \le 4m.\end{aligned}$$

We let $\theta_1 = \theta, \theta_2, \ldots, \theta_d$ denote the conjugates of θ and, simplifying our notation a bit, consider the product

$$N = \underbrace{\left(A_1^* + A_2^*\theta + \cdots + A_d^*\theta^{d-1}\right)}_{\text{primary factor}} \underbrace{\prod_{i=2}^{d} \left(A_1^* + A_2^*\theta_i + A_3^*\theta_i^2 + \cdots + A_d^*\theta_i^{d-1}\right)}_{\text{secondary factors}}. \tag{5.12}$$

This product is a nonzero rational integer since $\delta^{D_1+2D_2m} F(a^* + b^*\beta)$ is a nonzero algebraic integer, i.e., $N \neq 0$.

Since the argument leading to an upper bound for $|N|$ is so similar to the argument we used to conclude Gelfond's proof we will be brief. We estimate the absolute value of the *primary factor* through an application of the Maximum Modulus Principle; this estimate depends in a crucial way on the number of zeros of the function $F(z)$. The absolute value of each of the *secondary factors* is estimated through a simple application of the triangle inequality (given the above estimate for $\max\limits_{1\le j\le d}\{|A_j^*|\}$).

Estimating the primary factor. To estimate the primary factor in (5.12) we apply the Maximum Modulus Principle to the entire function

$$G(z) = \frac{\delta^{D_1+2D_2m^*}F(z)}{\prod_{a=1}^{m}\prod_{b=1}^{m}(z-(a+b\beta))} \ .$$

Given that $a \le m$ and $b \le m$ we know that if we take $R > m^*(1+|\beta|)$ then $|a^*+b^*\beta| < R$. Using such an R the Maximum Modulus Principle gives:

$$\begin{aligned}|\delta^{D_1+2D_2m^*}F(a^*+b^*\beta)| &\le |G|_R \prod_{a=1}^{m}\prod_{b=1}^{m}|(a^*-a)+(b^*-b)\beta| \\ &\le \frac{|\delta^{D_1+2D_2m^*}F|_R}{\left|\prod_{a=1}^{m}\prod_{b=1}^{m}(z-(a+b\beta))\right|_R}\prod_{a=1}^{m}\prod_{b=1}^{m}|(a^*-a)+(b^*-b)\beta| \ . \quad (5.13)\end{aligned}$$

It is easiest to estimate each of the factors in the right-hand side of the above inequality separately. We bound the numerator of the first factor, $|\delta^{D_1+2D_2m^*}F|_R$, through an application of the triangle inequality:

$$\begin{aligned}|\delta^{D_1+2D_2m^*}F|_R &= \left|\delta^{D_1+2D_2m^*}\sum_{k=0}^{D_1-1}\sum_{\ell=0}^{D_2-1}c_{k\ell}z^k\left(e^{\log\alpha z}\right)^\ell\right|_R \\ &\le \delta^{D_1+2D_2m^*}D_1D_2\max\{|c_{k\ell}|\}|z|_R^{D_1}\left|e^{\log\alpha z}\right|_R^{D_2} \\ &\le \delta^{D_1+2D_2m^*}D_1D_2c_0^{m^{3/2}\log m}R^{D_1}\left(e^{|\mathrm{Re}(\log\alpha)|R}\right)^{D_2} \ .\end{aligned}$$

In view of our choices of D_1 and D_2 and the fact that $R > m^*(1+|\beta|) > m$, we see that the previous inequality implies

$$|\delta^{D_1+2D_2m^*}F|_R < c_4^{m^{3/2}\log R+m^{1/2}R} \ .$$

The second factor in the above inequality (5.13) satisfies

$$\prod_{a=1}^{m}\prod_{b=1}^{m}|(a^*-a)+(b^*-b)\beta| \le \left(m^*(1+|\beta|)\right)^{m^2} < R^{m^2} \ .$$

We fix $R = (4m)^{3/2}(1+|\beta|)$ for the remainder of this argument. Then it is straightforward to produce a *lower* bound for the denominator in (5.13). Since for any z with $|z| = R$ we have

$$|z-(a+b\beta)| > (8m^{3/2}-m)(1+|\beta|) > m(1+|\beta|);$$

thus we obtain

$$\left|\prod_{a=1}^{m}\prod_{b=1}^{m}(z-(a+b\beta))\right|_R \ge (m(1+|\beta|))^{m^2}.$$

Putting all of these estimates together we have

$$|A_1^* + A_2^*\theta + A_3^*\theta^2 + \cdots + A_d^*\theta^{d-1}| \le c_4^{m^{3/2}\log R + m^{1/2}R} e^{m^2 \log R - m^2 \log(m(1+|\beta|)} \le c_5^{-(1/2)m^2 \log m}, \text{ by our choice of } R.$$

Estimating the secondary factors. As we indicated above, we estimate the secondary factors in N, the factors that involve one of the conjugates θ_i for $i = 2, 3, \ldots, d$, through the triangle inequality:

$$|A_1^* + A_2^*\theta_i + \cdots + A_d^*\theta_i^{d-1}| \le d \max_{1\le j\le d}\{|A_j^*|\} \max\{1, |\theta_i|\}^{d-1} \le dc_3^{m^{3/2}\log m} \max\{1, |\theta_1|, |\theta_2|, \ldots, |\theta_d|\}^{d-1} \le c_6^{m^{3/2}\log m}.$$

Therefore

$$\prod_{i=2}^{d} \left|A_1^* + A_2^*\theta_i + A_3^*\theta_i^2 + \cdots + A_d^*\theta_i^{d-1}\right| \le \left(c_6^{m^{3/2}\log m}\right)^{d-1} = \left(c_6^{d-1}\right)^{m^{3/2}\log m} \le c_7^{m^{3/2}\log m}.$$

We are led to the estimate:

$$0 < |N| < c_5^{-1/2m^2 \log m} c_7^{m^{3/2}\log m} \le c_8^{-1/2m^2 \log m} < 1,$$

for m sufficiently large. This contradiction completes our proof.

How to obtain the nonzero value

There are several ways to find an appropriate nonzero algebraic number. For historical accuracy we first consider Schneider's fairly complicated approach to this problem first.

Schneider's original method. To obtain his nonzero algebraic number Schneider considers several functions associated with the function $F(z)$, whose existence was guaranteed by the application of Siegel's Lemma. We will see that his reason for doing this is to find enough functions that, if they all vanish at the points under consideration, a certain Vandermonde determinant will vanish. We will see that this argument depends on an additional assumption about the algebraic nature of α. Specifically, Schneider points out that he may assume that α is not a root of unity. Indeed, if it is, then instead of considering the numbers $\alpha, \beta,$ and α^β at the very beginning of the proof one considers the numbers $\alpha^\beta, \beta^{-1},$ and α.

For notational simplicity in the argument below we retain the earlier notation $D_1 = [\sqrt{2d}\, m^{3/2}]$ and $D_2 = [\sqrt{2d}\, m^{1/2}]$. Using this notation Schneider defined

his associated functions as follows: for each $\sigma, 1 \le \sigma \le D_2$, let

$$F_\sigma(z) = \left[\prod_{1 \le a \le \sigma - 1, 1 \le b \le m} (z - (a + b\beta)) \right] F(z + \sigma - 1).$$

Notice that each $F_\sigma(z)$ vanishes at the prescribed zeros of $F(z)$, $z = a + b\beta, 1 \le a, b \le m$.

In order to understand the matrix Schneider introduces it is helpful to first rewrite the original auxiliary function as:

$$F(z) = P_{11}(z) + P_{12}(z)\alpha^z + P_{13}(z)\alpha^{2z} + \cdots + P_{1D_2}(z)\alpha^{(D_2-1)z}.$$

It is then an easy calculation to rewrite each of the functions $F_\sigma(z), 1 \le \sigma \le D_2$, in terms of polynomials $P_{11}, \ldots, P_{1D_2}$. If for each pair σ, τ we put

$$P_{\sigma\tau}(z) = \prod_{1 \le a \le \sigma - 1, 1 \le b \le m} (z - (a + b\beta)) P_{1\tau}(z + \sigma - 1),$$

then we have

$$F_\sigma(z) = \sum_{\tau=1}^{D_2} \alpha^{(\sigma-1)(\tau-1)} P_{\sigma\tau}(z) \alpha^{(\tau-1)z}. \tag{5.14}$$

We note that the vanishing of all of these functions at the indicated points translates into having a certain matrix product equal to zero. Specifically, for each $z = a + b\beta$

$$\begin{pmatrix} P_{11}(z) & P_{12}(z) & \cdots & P_{1D_2}(z) \\ P_{21}(z) & \alpha P_{22}(z) & \cdots & \alpha^{D_2-1} P_{2D_2}(z) \\ P_{31}(z) & \alpha^2 P_{32}(z) & \cdots & \alpha^{2(D_2-1)} P_{3D_2}(z) \\ \vdots & \vdots & \vdots & \vdots \\ P_{D_2 1}(z) & \alpha^{D_2-1} P_{D_2 2}(z) & \cdots & \alpha^{(D_2-1)(D_2-1)} P_{D_2 D_2}(z) \end{pmatrix} \times \begin{pmatrix} 1 \\ \alpha^z \\ \alpha^{2z} \\ \vdots \\ \alpha^{(D_2-1)z} \end{pmatrix} = \mathbf{0}.$$

By our application of Siegel's Lemma we know that not all of the polynomials in the first row of the above matrix are identically zero; we denote the nonzero polynomials in the first row by $P_{1\tau_1}(z), \ldots, P_{1\tau_r}(z)$ and consider the $r \times r$ matrix:

$$\begin{pmatrix} P_{1\tau_1}(z) & P_{1\tau_2}(z) & \cdots & P_{1\tau_r}(z) \\ \alpha^{(\tau_1-1)} P_{2\tau_1}(z) & \alpha^{(\tau_2-1)} P_{2\tau_2}(z) & \cdots & \alpha^{(\tau_r-1)} P_{2\tau_r}(z) \\ \alpha^{2(\tau_1-1)} P_{3\tau_1}(z) & \alpha^{2(\tau_2-1)} P_{3\tau_2}(z) & \cdots & \alpha^{2(\tau_r-1)} P_{3\tau_r}(z) \\ \vdots & \vdots & \vdots & \vdots \\ \alpha^{(r-1)(\tau_1-1)} P_{r\tau_1}(z) & \alpha^{(r-1)(\tau_2-1)} P_{r\tau_2}(z) & \cdots & \alpha^{(r-1)(\tau_r-1)} P_{r\tau_r}(z) \end{pmatrix}.$$

Following Schneider we temporarily let

$$\Pi_\sigma(z) = \prod_{1\le a\le \sigma-1, 1\le b\le m} (z-(a+b\beta))$$

so we have

$$P_{\sigma\tau}(z) = \Pi_\sigma(z) P_{1\tau}(z+\sigma-1).$$

Thus the above $r\times r$ matrix may be represented by the product:

$$\begin{pmatrix} P_{1\tau_1}(z) & P_{1\tau_2}(z) & \cdots & P_{1\tau_r}(z) \\ \Pi_2(z)P_{1\tau_1}(z-1) & \Pi_2(z)P_{1\tau_2}(z-1) & \cdots & \Pi_2(z)P_{1\tau_r}(z-1) \\ \Pi_3(z)P_{1\tau_1}(z-2) & \Pi_3(z)P_{1\tau_2}(z-2) & \cdots & \Pi_3(z)P_{1\tau_r}(z-2) \\ \vdots & \vdots & \vdots & \vdots \\ \Pi_r(z)P_{1\tau_1}(z-r+1) & \Pi_r(z)P_{1\tau_2}(z-r+1) & \cdots & \Pi_r(z)P_{1\tau_r}(z-r+1) \end{pmatrix}$$
$$\times \begin{pmatrix} 1 & \alpha^{\tau_1-1} & \cdots & \alpha^{(r-1)(\tau_1-1)} \\ 1 & \alpha^{\tau_2-1} & \cdots & \alpha^{(r-1)(\tau_2-1)} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \alpha^{\tau_r-1} & \cdots & \alpha^{(r-1)(\tau_r-1)} \end{pmatrix}$$

We denote the second matrix above by W and note that its determinant is Vandermonde. Then if we let $a_j x^{g_j}$ denote the leading coefficient of $P_{1\tau_j}(z)$ the determinant of the above product may be written as

$$\begin{aligned} &D(z) \\ &= \big(\Pi_1(z)\cdots\Pi_r(z)\big)\big(a_1\cdots a_r z^{g_1+\cdots+g_r}\times|W| + \text{ lower degree terms } \times|W|\big). \end{aligned}$$

If $D(z)$ vanishes identically then the coefficient of each power of z must equal zero. But the leading coefficient of $D(z)$ equals zero only if $|W|=0$, which would imply that α is a root of unity, contrary to our earlier assumption.

Schneider next shows that the function $D(z)$ is a polynomial in z, with coefficients involving powers of α. The degree of $D(z)$ may be shown to be less than $12m^2$ and so $D(z)$ has fewer than $12m^2$ zeros. Thus there exists a pair $a^*+b^*\beta, 1\le a^*, b^* < 4m$ so that $D(a^*+b^*\beta)\neq 0$. This means that none of the rows of the above matrix can vanish at $a^*+b^*\beta$ and, looking at the first row, we deduce that $F(a^*+b^*\beta)\neq 0$.

An alternate way to obtain the nonzero value. We saw above that Schneider used a subtle argument, based on the nonvanishing of a Vandermonde determinant, to obtain a point $a^*+b^*\beta$ which produced a nonzero algebraic number $F(a^*+b^*\beta)$ that eventually lead to a positive integer less than 1. It is apparent that obtaining such a nonzero algebraic number was central to both Gelfond's and Schneider's methods. Perhaps not unexpectedly, finding alternate approaches to finding a nonzero value for large classes of analytic functions became an important area of research in transcendental number theory in the second half of the twentieth century. We conclude this chapter with another ap-

proach to obtaining the all-important nonzero value for a so-called exponential polynomial. (We will use the proposition below in the last chapter.)

This approach is based on providing a count of the total number of zeros an exponential polynomial can have. What is surprising about this approach is that, at least in the real case, the proof requires no ideas beyond basic calculus. (This proposition, due to Polya [23], was used by Gelfond in 1962 when he provided a simpler proof of his theorem in case both α and β are real.)

Proposition 5.4. *Let $P_1(z), P_2(z), \ldots, P_k(z)$ be nonzero polynomials with real coefficients and degrees $d_1, d_2, \ldots, d_k$, respectively. Suppose $\omega_1, \omega_2, \ldots, \omega_k$ are distinct real numbers. Then the function*

$$F(z) = P_1(z)e^{\omega_1 z} + P_2(z)e^{\omega_2 z} + \cdots + P_k(z)e^{\omega_k z}$$

has at most

$$d_1 + d_2 + \cdots + d_k + k - 1$$

real zeros.

Proof. The proof of the proposition is by induction on $n = d_1 + d_2 + \cdots + d_k + k$. If $n = 1$ then $k = 1$ and $d_1 = 0$. Thus $F(z) = a_1 e^{\omega_1 z}$. Since $a_1 \neq 0$, $F(z)$ has no zeros.

We now take $m \geq 2$, assume the result has been established for all functions with $n = d_1 + d_2 + \cdots + d_k + k < m$, and let $F(z)$ be a function, as above, with $d_1 + d_2 + \cdots + d_k + k = m$. Let N denote the number of real zeros of $F(z)$. We note that, after multiplying $F(z)$ by $e^{-\omega_k z}$, we may assume that $\omega_k = 0$. The trick is to next apply Rolle's Theorem, by which we know that the number of zeros of

$$\begin{aligned} F'(z) &= \omega_1 P_1(z)e^{\omega_1 z} + P_1'(z)e^{\omega_1 z} + \omega_2 P_2(z)e^{\omega_2 z} + P_2'(z)e^{\omega_2 z} + \cdots + P_k'(z) \\ &= (\omega_1 P_1(z) + P_1'(z))\, e^{\omega_1 z} + (\omega_2 P_2(z) + P_2'(z))\, e^{\omega_2 z} + \cdots + P_k'(z), \end{aligned}$$

is at least $N - 1$.

Notice that in the above representation of $F'(z)$ we have $\deg(\omega_j P_j(z) + P_j'(z)) \leq d_j$ for each $j = 1, \ldots, k-1$. However the degree of the coefficient of the term e^0 is one less than the degree of the coefficient of e^0 in $F(z)$. Therefore we may apply the induction hypothesis to conclude that

$$N - 1 \leq \text{number of zeros of } F'(z) \leq d_1 + d_2 + \cdots + d_k + k - 2,$$

from which the proposition follows.

If we assume that $\alpha, \log\alpha$, and β are all real, we can apply the above proposition to establish the existence of the nonzero value needed to deduce the Gelfond-Schneider Theorem from Proposition 5.1. We recall the function $F(z)$ from (5.1), which we rewrite as:

$$F(z) = \sum_{k=0}^{D_1-1} \sum_{\ell=0}^{D_2-1} c_{k\ell} z^k \alpha^{\ell z} = \sum_{\ell=0}^{D_2-1} \left(\sum_{k=0}^{D_1-1} c_{k\ell} z^k \right) e^{(\ell \log\alpha) z}. \tag{5.15}$$

The exponents $\omega_\ell = \ell \log \alpha,\ 0 \le \ell \le D_2 - 1$, are distinct real numbers. Therefore the number of real zeros of the function (5.15) is less than:

$$\underbrace{(D_1 - 1) + \cdots + (D_1 - 1)}_{D_2 \text{terms}} + D_2 - 1 = D_1 D_2 - 1.$$

Recalling the notation used in Schneider's proof, $D_1 = [\sqrt{2d}m^{3/2}]$ and $D_2 = [\sqrt{2d}m^{1/2}]$, we see that the function $F(z)$ has a nonzero value of the form $a^* + b^*\beta$ with $a^*, b^* \le 2dm$.

Exercises

1. Verify that the estimate in the last line of this chapter, $a^*, b^* \le 2dm$, may be substituted for Schneider's estimate to complete the proof of Proposition 5.1 in the real case.

2. Prove Lemma 5.3.

3. Show that the constant c_1 in (5.9) may be taken to equal the explicit quantity:

$$2d^2 H(\theta)^d (1 + H(p_\beta)) H(p_\alpha) H(p_{\alpha^\beta}).$$

4. Verify the validity of the expression (5.14).

5. Suppose that $F(z)$ is a nonzero entire function that satisfies $|F|_R \le e^{R^\kappa}$ for all sufficiently large R. Show that F cannot have more than $R^{\kappa+\epsilon}$ zeros in the disks of all complex numbers z satisfying $|z| \le R$ as R approaches infinity.

Chapter 6
Hilbert's seventh problem and transcendental functions

So far we have not said much about an important portion of Hilbert's seventh problem, wherein he said

we expect transcendental functions to assume, in general, transcendental values for ... algebraic arguments ... we shall still consider it highly probable that the exponential function $e^{i\pi z}$ ... will ... always take transcendental values for irrational algebraic values of the argument z

In other words, Hilbert speculated that if $f(z)$ is a transcendental function and if α is an irrational algebraic number then $f(\alpha)$ should be a transcendental number.

The function $e^{i\pi z}$, which Hilbert explicitly mentions, is covered by the Gelfond-Schneider Theorem because $e^{i\pi} = -1$ is an allowable value of α. A simple question is: For which numbers γ do we already know that the function $e^{\gamma z}$ will, in Hilbert's words, *always take transcendental values for irrational algebraic values of the argument z*. The partial answer we can already give to this question comes in two parts. The first part preceded Hilbert's lecture. The Hermite-Lindemann Theorem established the transcendence of e^{α} for any nonzero algebraic number α. It follows, of course, that if in the original question we take γ to be any nonzero algebraic number then the function $e^{\gamma z}$ certainly takes on transcendental values for any nonzero algebraic values of the argument z. The second part of our answer to this question comes from the Gelfond-Schneider Theorem. If γ is the natural logarithm of any algebraic number $\alpha \neq 1$ then the function $e^{\gamma z} = \alpha^z$ also takes on transcendental values for any irrational algebraic values of the argument z. Thus we have the partial answer to the original question: For any $\gamma \in \{\alpha, \log \alpha : \alpha \text{ an algebraic number different from 0 or 1}\}$ the function $e^{\gamma z}$ takes on transcendental values for any irrational algebraic values of the argument z. We note that although $\gamma = i\pi$ is in the above set of values it is, unfortunately, still a fairly small set of values.

The disclaimer *in general* in Hilbert's posing of his problem that transcendental functions should take transcendental values at irrational algebraic numbers saved him from possible embarrassment when counter-examples to the most general interpretation of this conjecture were given. We will not consider

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this topic here but notice that it is easy to exhibit a transcendental function that is algebraic at any number of prescribed algebraic numbers. For example, if $\alpha_1, \alpha_2, \ldots, \alpha_\ell$ are algebraic numbers, then

$$f(z) = e^{(z-\alpha_1)(z-\alpha_2)\cdots(z-\alpha_\ell)} \tag{6.1}$$

produces an algebraic value for each of $z = \alpha_1, \alpha_2, \ldots, \alpha_\ell$.

Despite the simplicity of the above counterexample to one interpretation of Hilbert's conjecture, a slight generalization of the question posed by Hilbert has proven to be a fruitful one for transcendental number theory. Instead of considering the values of a single function it is natural to consider the values of two functions simultaneously. The reason this is a natural generalization of Hilbert's question is because this point of view is already implicit in that question. Asking whether a transcendental function $f(z)$ takes on transcendental values for algebraic values of the argument is equivalent to asking: Can the functions $f(z)$ and z be simultaneously algebraic? (Or, more precisely, can the functions $f(z)$ and z be simultaneously algebraic, possibly with a finite number of exceptions?) The Hermite-Lindemann Theorem says that e^z and z are not simultaneously algebraic except when $z = 0$. The Gelfond-Schneider Theorem has two possible statements in terms of two functions and their values. One says that for an algebraic number $\alpha \neq 0, 1$ the functions z and α^z are not simultaneously algebraic except when z is a rational number; another says that if β is an irrational algebraic number then the functions e^z and $e^{\beta z}$ cannot be simultaneously algebraic except when $z = 0$.

In this chapter we consider the question of when two algebraically independent functions can be simultaneously algebraic, and see that some important, but far from definitive, steps have been taken towards answering this question. Taking z as one of the functions under consideration allows us to restate what might be called *classical* transcendence theorems. To move into the modern era we want to expand the type of functions under consideration beyond the functions z and e^z to the Weierstrass $\wp$-function. We first consider two, algebraically independent exponential functions and give a theorem that straddles the line between *classical* and *modern* transcendental number theory—the *Six Exponentials Theorem*. This result is classical in that its proof is simply an elaboration of Schneider's proof of the Gelfond–Schneider Theorem and yet it is modern in that it examines the transcendence of values of functions rather than of particular numbers.

Theorem 6.1. *Let $\{x_1, x_2\}$ and $\{y_1, y_2, y_3\}$ be two $\mathbf{Q}$-linearly independent sets of complex numbers. Then at least one of the six numbers*

$$e^{x_1y_1}, \quad e^{x_1y_2}, \quad e^{x_1y_3}, \quad e^{x_2y_1}, \quad e^{x_2y_2}, \quad e^{x_2y_3}$$

is transcendental.

For example, if we consider the sets $\{1, e\}$ and $\{e, e^2, e^3\}$, then the Six Exponentials Theorem implies that at least one of the following numbers is

transcendental:

$$e^e, e^{e^2}, e^{e^3}, e^{e^4} \ .$$

The Six Exponentials Theorem is easily restated as a result concerning the values of two functions:

Restatement of the Six Exponentials Theorem. *Two algebraically independent exponential functions, $e^{x_1 z}$ and $e^{x_2 z}$ cannot be simultaneously algebraic at three $\mathbf{Q}$-linearly independent complex numbers y_1, y_2, and y_3.*

Sketch of proof. The proof of the Six Exponentials Theorem closely parallels Schneider's solution of Hilbert's seventh problem, so we will be brief.

Step 1. Assume that all of the values $e^{x_i y_j}$ are algebraic. Thus for any $P(x, y) \in \mathbf{Z}[x, y]$, we notice that the values of the function $F(z) = P(e^{x_1 z}, e^{x_2 z})$ will be algebraic when evaluated at y_1, y_2, y_3, or any $\mathbf{Z}-$linear combination of them. That is, for any integers k_1, k_2, and k_3, the quantity $F(k_1 y_1 + k_2 y_2 + k_3 y_3)$ is an algebraic number.

Step 2. Apply Siegel's Lemma to find a nonzero integral polynomial

$$P(x, y) = \sum_{m=0}^{D_1 - 1} \sum_{n=0}^{D_2 - 1} a_{mn} x^m y^n \ ,$$

having "modestly sized" integral coefficients, such that if we let

$$F(z) = P(e^{x_1 z}, e^{x_2 z}) \ ,$$

then $F(z) = 0$ for all $z \in \{k_1 y_1 + k_2 y_2 + k_3 y_3 : 0 \leq k_j < K\}$. Before proceeding to the next step, we note that since the two functions $e^{x_1 z}$ and $e^{x_2 z}$, which we compose with $P(x, y)$ in order to produce $F(z)$, are so similar, it is natural to take take $D_1 = D_2$.

Step 3. As we have seen there are several ways to obtain a nonzero value that will, if everything is set up correctly, lead to a contradictory nonzero integer. In this proof we use a zeros estimate based on the observation that $F(z)$ is not identically zero. Specifically, the following lemma, whose proof is left as an exercise, ensures that the function $F(z)$ has an nonzero value that may be exploited to conclude the proof.

Lemma 6.2. *There exists a positive integer M such that*

$$F(k_1 y_1 + k_2 y_2 + k_3 y_3) = 0 \ ,$$

for all $0 \leq k_j < M$, while there exists some triple k_1^*, k_2^*, k_3^*, satisfying $0 \leq k_j^* \leq M$, such that

$$F(k_1^* y_1 + k_2^* y_2 + k_3^* y_3) \neq 0 \ .$$

Step 4. It is possible to use the nonzero algebraic number $F(k_1^*y_1+k_2^*y_2+k_3^*y_3)$ to obtain a nonzero integer whose absolute value is less than 1.

The Schneider-Lang Theorem

In this section we consider a conjecture which is an natural analogue of Hilbert's, albeit for two functions. This conjecture captures the essence of Hermite's result, the Hermite-Lindemann Theorem, and the Gelfond-Schneider Theorem.

First Conjecture. *Two algebraically independent functions should not be simultaneously algebraic at a point, unless there is some special reason. That is, if $f(z)$ and $g(z)$ are algebraically independent functions, then for just about any $z_0 \in \mathbf{C}$, at least one of the values $f(z_0)$ or $g(z_0)$ should be a transcendental number.*

Our earlier example of the function $f(z) = e^{(z-\alpha_1)(z-\alpha_2)\cdots(z-\alpha_\ell)}$, which is algebraically independent of the function $g(z) = z$, shows the need for the phrase "unless there is some special reason" in the above conjecture.

Refined Conjecture. *Two algebraically independent functions cannot simultaneously be algebraic at very many different complex numbers.*

Both our reframing of the Gelfond-Schneider Theorem, in Exercise 1 (below), and the Six Exponentials Theorem point to the linear independence of the points under consideration as being a reasonable hypothesis. While this point of view remains an important one, there is another point of view that leads to an important result. This point of view is to refine the phrase "simultaneously algebraic" in the above refined conjecture.

Before we examine this portion of the refined conjecture we point out that the example we gave above indicates that the number of points at which the two functions are simultaneously algebraic must depend on some specific properties of the functions themselves. Specifically, if $P(z)$ is any nonzero polynomial with rational coefficients, of degree $d \geq 1$, then the algebraically independent functions

$$f(z) = z \quad \text{and} \quad g(z) = e^{P(z)}$$

are simultaneously algebraic at the d zeros of $P(z)$. So in this example "too many" must be connected with the degree of $P(z)$.

What cannot be taken away from the above example is how the degree of $P(z)$ plays a role in determining an upper bound on the number of simultaneous algebraic points for z and $e^{P(z)}$. There is a clue in the exercises at the end of Chapter 5 where you were invited to obtain a zeros estimate based on the order of growth of the function. That result establishes that if a nonzero entire function $F(z)$ satisfies $|F|_R \leq e^{R^\kappa}$ for all sufficiently large R, then F cannot have more than $R^{\kappa+\epsilon}$ zeros in the disks of all complex numbers z satisfying $|z| \leq R$ as R approaches infinity. This result implies that for any particular

complex number β, such a nonconstant entire function $F(z)$ satisfies

$$\text{card}\Big\{ z \in \mathbf{C} : F(z) = \beta \text{ with } |z| \leq R \Big\} < R^{\kappa+\epsilon} \ .$$

So the number of times such an entire function can attain any particular algebraic value is bounded by a function of R and κ. Extending this observation, it is reasonable to imagine that κ might influence how many times a particular entire function takes values in any fixed *set* of algebraic numbers, or more generally in any finite extension K of $\mathbf{Q}$.

We have never formally described the exponents κ above. To do so we say that an entire function $F(z)$ has *order of growth* less than or equal to κ if for all $\epsilon > 0$,

$$|f(z)| < e^{|z|^{\kappa+\epsilon}} \text{ for } |z| \text{ sufficiently large.}$$

When considering the simultaneous algebraic values of two algebraically independent functions, the orders of growth of the two functions will play a role. Surprisingly, however, we will see that if we wish to give an upper bound for the number of points in a disk at which two algebraically independent functions simultaneously take values from a prescribed collection of algebraic numbers, neither the radius of the disk nor the cardinality of the set of algebraic values appears in our bound. Instead, the upper bound depends only on the degree of the field extension of $\mathbf{Q}$ containing the given set of algebraic numbers and the orders of growth of the functions.

We now state an important result due to Serge Lang (1964) [18] known as the Schneider-Lang Theorem. In 1949 Schneider [26] proved two general theorems concerning two algebraically independent functions being simultaneously algebraic, but Lang's formulation is particularly succinct. This result concerns the values of meromorphic functions, rather than entire functions, and its statement requires the order of growth of such a function $f(z)$. This order can be defined in several ways; one is given in the exercises.

Theorem 6.3 (Schneider-Lang Theorem). *Let $f_1(z)$ and $f_2(z)$ denote two algebraically independent meromorphic functions with finite orders of growth ρ_1 and ρ_2, respectively. Suppose $f_1(z)$ and $f_2(z)$ satisfy polynomial algebraic differential equations over a number field F; that is, there exists a finite collection of functions $f_3(z), f_4(z), \ldots, f_n(z)$ such that the differential operator $\frac{d}{dz}$ maps the ring $F[f_1(z), f_2(z), \ldots, f_n(z)]$ into itself. Then for any number field E containing F,*

$$\text{card}\Big\{ z \in \mathbf{C} : f_1(z) \in E, \ldots, \ f_n(z) \in E \Big\} \leq (\rho_1 + \rho_2)[E : \mathbf{Q}] \ .$$

Before outlining the proof of this result, let's see how the Schneider-Lang Theorem can be applied to obtain the Gelfond-Schneider Theorem. We begin by assuming that each of α, β and α^β is algebraic and let E be the number field given by $E = \mathbf{Q}\left(\alpha, \beta, \alpha^\beta\right)$. We put $f_1(z) = e^z$ and $f_2(z) = e^{\beta z}$. Since β is irrational, we know that $f_1(z)$ and $f_2(z)$ are algebraically independent functions; we also know they each have order of growth 1. Moreover, each of

these functions satisfies an algebraic differential equation,

$$\frac{dy}{dz} = y \quad \text{and} \quad \frac{dy}{dz} = \beta y \ ,$$

respectively. So if we take $K = \mathbf{Q}(\beta) \subseteq E$, then we see that $K[f_1(z), f_2(z)]$ is closed under differentiation. Thus we may apply the Schneider-Lang Theorem and deduce that there are only *finitely* many points $z \in \mathbf{C}$ such that $f_1(z) \in E$ and $f_2(z) \in E$. However, we note that for all integers k,

$$f_1(k \log \alpha) = \alpha^k \in E \quad \text{and} \quad f_2(k \log \alpha) = \left(\alpha^\beta\right)^k \in E \ ,$$

which contradicts the previous sentence. Therefore we conclude that if α and β are algebraic, then α^β is transcendental.

A sketch of the proof of the Schneider-Lang Theorem. We prove the Schneider-Lang Theorem under the assumption that f_1 and f_2 are entire. (If they are meromorphic we can reduce the proof to this case by multiplying each of the meromorphic functions by the appropriate entire function that vanishes at its poles.)

We fix a number field E and let

$$\Omega = \{z \in \mathbf{C} : f_1(z) \in E, \ f_2(z) \in E, \ldots, f_n(z) \in E\} \ .$$

Our aim is to show that $\text{card}(\Omega) \leq (\rho_1 + \rho_2)[E : \mathbf{Q}]$. We establish this by assuming that $\{w_1, w_2, \ldots, w_\ell\}$ is a set of distinct elements from Ω and showing that ℓ cannot be too large.

An application of Siegel's Lemma allows us to solve a system of linear equations to find a nonzero polynomial $P(x, y)$ such that the function $F(z) = P(f_1(z), f_2(z))$ vanishes at each of the points $w_1, w_2, \ldots, w_\ell$, each with some multiplicity. Since $f_1(z)$ and $f_2(z)$ are algebraically independent functions, $F(z)$ is not identically zero. Thus we let t_0 be the smallest positive integer such that there exists an index l_0, $1 \leq l_0 \leq \ell$, satisfying

$$\frac{d^{t_0}}{dz^{t_0}} F(w_{l_0}) \neq 0 \ .$$

If we let $\Gamma = \frac{d^{t_0}}{dz^{t_0}} F(w_{l_0})$, then by the hypothesis it follows that there exists a polynomial $P(x_1, x_2, \ldots, x_n)$ with coefficients in E such that

$$\Gamma = P(f_1(w_{l_0}), f_2(w_{l_0}), \ldots, f_n(w_{l_0})).$$

Thus we conclude that Γ is an algebraic number in E. We henceforth assume Γ is an algebraic integer.

Then if we let $\Gamma_1 (= \Gamma), \Gamma_2, \ldots, \Gamma_d$ denote the conjugates of Γ we know that

$$N = \Gamma_1 \times \Gamma_2 \times \cdots \times \Gamma_d,$$

is a rational integer.

It is relatively straightforward to estimate each of the values $|\Gamma_k|, k = 2, \ldots, d$. As we have before, we may employ the Maximum Modulus Principle to estimate $|\Gamma_1|$. Specifically, we consider the function

$$G(z) = \frac{F(z)}{\prod_{l=1}^{\ell}(z - w_l)^{t_0 - 1}}$$

on a disk of radius $t_0^{1/(\rho_1+\rho_2)}$ to get an upper bound for $|\Gamma|$ involving ρ_1, ρ_2, and $\deg(\Gamma)$. Applying our bounds it is possible to show that if $\ell > (\rho_1 + \rho_2)[E : \mathbf{Q}]$, then the integer N satisfies $0 < |N| < 1$. This contradiction establishes the result.

Elliptic Functions

We end this chapter with arguably the second most important function in number theory, after the usual exponential function e^z, the meromorphic Weierstrass $\wp$-function. Just as there are several characterizations of e^z, there are several characterizations of $\wp(z)$. We will use these characterizations (properties) in establishing transcendence results associated with $\wp(z)$, and, given what we have already seen, they are not surprisingly both analytic and algebraic in nature.

A series representation for $\wp(z)$. For any nonzero $w \in \mathbf{C}$, we know that there exists an entire function that is periodic with respect to $\mathbf{Z}w$; namely the function $f(z) = e^{\frac{2\pi i}{w}z}$. A critical difference between the function $\wp(z)$ and e^z is that $\wp(z)$ has *two* $\mathbf{Q}$-linearly independent periods (such a function is said to be *doubly periodic*). Moreover, just as there is an exponential function periodic with respect to $\mathbf{Z}w$ for any nonzero $w \in \mathbf{C}$, given any two $\mathbf{Q}$-linearly independent complex numbers w_1 and w_2 satisfying $w_2/w_1 \notin \mathbf{R}$, there exists a Weierstrass $\wp$-function that is periodic with respect to the lattice $W = \mathbf{Z}w_1 + \mathbf{Z}w_2 \subseteq \mathbf{C}$.

Liouville demonstrated that an entire, bounded function must be a constant. It follows that the only doubly periodic functions that can be represented by an everywhere-convergent power series are the constant functions. Thus there cannot exist as attractive a power series for $\wp(z)$ as there is for e^z. However, the complex numbers for which any non-constant, doubly periodic function is not defined must form a *discrete* subset in the complex plane. The Weierstrass $\wp$-function is normalized so that the points at which it is not defined are precisely its periods. Moreover, the behavior of $\wp(z)$ at the periods $w \in W$ will be well understood–we will see that it has essentially the same behavior as the function $1/(z - w)^2$ for z near w.

We do not develop this theory here but it more-or-less follows from the above brief discussion that the Weierstrass $\wp$-function is represented by a series of the form:

$$\wp(z) = \frac{1}{z^2} + \sum_{w \in W'} \left(\frac{1}{(z - w)^2} - \frac{1}{w^2} \right),$$

where W' denotes the nonzero elements of $W = \mathbf{Z}w_1 + \mathbf{Z}w_2$.

Although it is not immediately obvious from the above series representation, $\wp(z)$ is indeed a periodic function with respect to the lattice W. (This assertion can be established through a simple trick. Define two new functions by

$$f_1(z) = \wp(z + w_1) - \wp(z) \text{ and } \quad f_2(z) = \wp(z + w_2) - \wp(z) \ ,$$

where $W = \mathbf{Z}w_1 + \mathbf{Z}w_2$. Then $f_1(z)$ and $f_2(z)$ are meromorphic functions whose derivatives are identically zero; thus, they are constant functions. Evaluating $f_1(-w_1/2)$ and $f_2(-w_2/2)$, and using the easily observed fact that $\wp(z)$ is an even function demonstrates that these functions are identically zero.)

The derivative of $\wp(z)$. The first step in uncovering a relationship between $\wp(z)$ and $\wp'(z)$ is to look at the Laurent series of $\wp(z)$ centered at $z = 0$. Since $\wp(z)$ is an even function we can deduce that the coefficients of the odd powers of z must all equal 0. Thus we may express the Laurent series for $\wp(z)$ about $z = 0$ as

$$\wp(z) = \frac{1}{z^2} + c_0 + c_2 z^2 + c_4 z^4 + \cdots \ .$$

But we also know that

$$\wp(z) - \frac{1}{z^2} = \sum_{w \in W'} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2} \right) \ ,$$

and the right-hand side of the above identity vanishes at $z = 0$, so we may conclude that $c_0 = 0$. Thus

$$\wp(z) = \frac{1}{z^2} + c_2 z^2 + c_4 z^4 + \cdots \ ,$$

formal differentiation of which yields the convergent series

$$\wp'(z) = -\frac{2}{z^3} + 2c_2 z + 4c_4 z^3 + \cdots \ .$$

(Note that $\wp'(z) = -\sum_{w \in W} \frac{2}{(z-w)^3}$ so $\wp'(z)$ is also doubly periodic with respect to W.)

It is a fairly difficult exercise in most graduate complex variables courses to show that it follows from the above two expressions that for all complex numbers $z \notin W$,

$$\wp'(z)^2 = 4\wp(z)^3 - g_2 \wp(z) - g_3 \ ,$$

where g_2 and g_3 have the explicit representations

$$g_2 = 20c_2 = 60 \sum_{w \in W'} \frac{1}{w^4} \qquad g_3 = 28c_4 = 140 \sum_{w \in W'} \frac{1}{w^6}.$$

It is part of the theory of that the polynomial $4x^3 - g_2 x - g_3$ has distinct roots.

An application of the Schneider-Lang Theorem to $\wp(z)$. With this brief introduction we are already in a position to deduce transcendence results about

numbers associated with a Weierstrass $\wp$-function from the Schneider-Lang Theorem. Many of these results were established by Schneider in 1934 [25], before the formalization of the Schneider-Lang Theorem. We first consider Schneider's elliptic analogue of Lindemann's Theorem.

Theorem 6.4. *Suppose that the coefficients g_2 and g_3 of the differential equation*

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3 \ ,$$

are algebraic. Let W denote the lattice of periods for $\wp(z)$. Then every nonzero element of W is transcendental.

(We note for historical accuracy that in 1932 Siegel [28] proved that if the coefficients g_2 and g_3 of the differential equation are algebraic, and $W = \mathbf{Z}w_1 + \mathbf{Z}w_2$, then either w_1 or w_2 is transcendental.)

Our deduction of the above theorem from the Schneider-Lang Theorem requires the following elementary lemma.

Lemma 6.5. *Suppose that we factor the polynomial differential equation of $\wp(z)$ over the complex numbers as*

$$\wp'(z)^2 = 4\wp^3(z) - g_2\wp(z) - g_3 = 4(\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3). \quad (6.2)$$

If we write the period lattice for $\wp(z)$ as $W = \mathbf{Z}w_1 + \mathbf{Z}w_2$, then reordering e_1, e_2, and e_3, if necessary, it follows that

$$\wp\left(\frac{w_1}{2}\right) = e_1 \ , \quad \wp\left(\frac{w_2}{2}\right) = e_2 \ , \quad \textit{and} \quad \wp\left(\frac{w_1 + w_2}{2}\right) = e_3 \ .$$

Proof. For simplicity let $w_3 = w_1 + w_2$. In view of the factorization (6.2), we need only show that for each $n = 1, 2, 3$, $\wp'(\frac{w_n}{2}) = 0$. Since $\wp(z)$ is even, $\wp'(z)$ is an odd function, and thus, using the fact that $w_n \in W$,

$$\wp'\left(\frac{w_n}{2}\right) = \wp'\left(\frac{w_n}{2} - w_n\right) = \wp'\left(-\frac{w_n}{2}\right) = -\wp'\left(\frac{w_n}{2}\right) \ ,$$

from which the lemma follows.

The proof of the transcendence of the nonzero periods of a Weierstrass $\wp$-function with algebraic g_2 and g_3 then follows from applying the Schneider-Lang Theorem to the functions $f_1(z) = z$ and $f_2(z) = \wp(z)$ at the points $w/2, w/2 + w, w/2 + 2w \ldots$. The one complication is that $w/2$ might also be in W, and so be a pole of $\wp(z)$ (in other words it is very well possible that $w = mw_1 + nw_2$ where both m and n are even). If this is the case divide w by a sufficiently large power of 2, $\frac{w}{2^k} = m'w_1 + n'w_2$, so that not both m' and n' even. Then let $w' = m'w_1 + n'w_2$ and consider z and $\wp(z)$ at the points $w'/2, w'/2 + w', w'/2 + 2w', \ldots$. The transcendence of w follows from the transcendence of w'.

It is possible to deduce results about values of the classical Gamma- and Beta-functions from Theorem 6.4. We restrict ourselves to a single such result.

Corollary 6.6. *The real number*

$$\frac{\Gamma(1/4)^2}{\sqrt{\pi}}$$

is transcendental.

Sketch of proof. Let

$$I = \int_1^\infty \frac{dx}{\sqrt{4x^3 - 4x}}. \tag{6.3}$$

First show that the number $2I$ is a nonzero period of the Weierstrass $\wp$-function that satisfies the differential equation $y' = 4y^3 - 4y$. Therefore, I represents a transcendental number. Then, using the change of variables $x = \frac{1}{\sqrt{u}}$ show that

$$I = -\frac{1}{2}\int_1^0 \frac{u^{-\frac{3}{2}}\,du}{\sqrt{u^{-\frac{3}{2}} - u^{-\frac{1}{2}}}} = \frac{1}{2}\int_0^1 u^{-\frac{3}{4}}(1-u)^{-\frac{1}{2}}\,du\ .$$

Using the standard identities involving the Gamma-function:
First identity. For positive real numbers a and b,

$$\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} = \int_0^1 x^{a-1}(1-x)^{b-1}\,dx\ , \tag{6.4}$$

and,
Second identity. For a complex number z for which neither z nor $1 - z$ is a negative integer,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}. \tag{6.5}$$

Conclude that

$$I = \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{2\Gamma(\frac{3}{4})} = \frac{1}{2\sqrt{2}}\frac{\Gamma(\frac{1}{4})^2}{\sqrt{\pi}}\ .$$

The corollary follows.

Additional remarks about $\wp(z)$. The few properties of the $\wp$-function we have seen so far are not sufficient for us to deduce from the Schneider-Lang Theorem elliptic analogues of either the Hermite-Lindemann or Gelfond-Schneider theorems. We are missing two things. The first is a $\wp$-version of the *addition formula* $e^{x+y} = e^x e^y$ that holds for the usual exponential function. As one might expect the analogous formula for $\wp(z)$ is a more complicated matter. Indeed, it is given by the formula:

$$\wp(z_1 + z_2) = -\wp(z_1) - \wp(z_2) + \frac{1}{4}\Big(\frac{\wp'(z_2) - \wp'(z_1)}{\wp(z_2) - \wp(z_1)}\Big)^2. \tag{6.6}$$

The verification of this formula is an application of Liouville's result that a bounded, entire function must be constant.

The second missing piece of mathematics is an elliptic version of the algebraic independence of e^z and $e^{\beta z}$ when β is an irrational number. We do not propose to develop this theory here but only report that if β is a complex number then $\wp(z)$ and $\wp(\beta z)$ are algebraically independent if and only if $rank_{\mathbf{Z}} W \cap \frac{1}{\beta} W \neq 2$.

Theorem 6.7 (Elliptic version of the Hermite-Lindemann Theorem). Suppose α is a nonzero algebraic number and that $\wp(z)$ has algebraic g_2 and g_3. Then $\wp(\alpha)$ is transcendental.

Proof. Apply the Schneider-Lang Theorem to the functions z and $\wp(z)$ at the points $\alpha, 2\alpha, 3\alpha, \ldots$.

Theorem 6.8 (Elliptic version of the Gelfond-Schneider Theorem). Suppose $\wp(z)$ has algebraic g_2 and g_3. Suppose further that β is an algebraic number so that the functions $\wp(z)$ and $\wp(\beta z)$ are algebraically independent. If $\wp(u)$ is algebraic then $\wp(\beta u)$ is transcendental.

Proof. A thought-provoking exercise.

Exercises

1. Establish the following corollary of the Gelfond-Schneider Theorem that involves two functions at two points: *Given an irrational $\xi \in \mathbf{C}$, the two functions $e^{\xi z}$ and z cannot be simultaneously algebraic at two $\mathbf{Q}$-linearly independent complex numbers x_1 and x_2.*

2. Suppose $F(z)$ is a nonzero, meromorphic function. Suppose y_1, y_2, y_3 are $\mathbf{Q}$-linearly independent complex numbers. Show that there exists a positive integer M such that

$$F(k_1 y_1 + k_2 y_2 + k_3 y_3) = 0 \ ,$$

for all $0 \leq k_j < M$, while there exists some triple k_1^*, k_2^*, k_3^*, satisfying $0 \leq k_j^* \leq M$, where

$$F(k_1^* y_1 + k_2^* y_2 + k_3^* y_3) \neq 0 \ .$$

3. In this exercise you will be asked to deduce the addition formula (6.6). Fix a $y \notin W$ and define the function $f(z)$ by

$$f(z) = \wp(z+y) + \wp(z) - \frac{1}{4}\Big(\frac{\wp'(z) - \wp'(y)}{\wp(z) - \wp(y)}\Big)^2 \ .$$

First, show that the function

$$\frac{\wp'(z) - \wp'(y)}{\wp(z) - \wp(y)}$$

has a pole of order 1 at any element of the set

$$W_y = \{w, w+y, w-y : w \in W\} \ .$$

Then conclude that $f(z)$ does not have a pole at any point in W_y. Second, show that $f(z)$ is a bounded entire function, and letting $y \to 0$, conclude that $f(z) = -\wp(y)$.

4. Suppose $\wp(z)$ is an elliptic function with both g_2 and g_3 being algebraic. Let w be a nonzero period of $\wp(z)$. Show that both e^w and $\wp(i\pi)$ are transcendental.

5. Prove the zeros estimate used in the sketch of the proof of the Schneider-Lang Theorem.

6. A meromorphic function f has a finite order of growth if there exists a number κ so that for any $\epsilon > 0$

$$\max\{|f(z)| : |z| = R\} \leq e^{R^{\kappa+\epsilon}} \ ,$$

for all sufficiently large values of R that avoid the poles of $f(z)$. The infimum of all such κ is the order of growth of the function. Show that using this definition, the order of growth of a meromorphic function $f(z)$ equals

$$\limsup_{R\to\infty} \frac{\log\log \max\{|f(z)| : |z| = R\}}{\log R} \ .$$

Chapter 7
Variants and generalizations

There are several variants and/or extensions of the original Gelfond-Schneider Theorem that led to many significant advances in transcendental number theory in the twentieth century. Perhaps the four themes that have led to the greatest advancements are:

1. Changing or slightly altering the arithmetic nature of the numbers α and β.
2. Providing quantitative versions of the original Gelfond-Schneider Theorem and its generalizations.
3. Giving analogues in other settings.
4. Moving beyond the transcendence of a single number to the algebraic independence of two or more numbers.

In this short chapter we briefly touch on the first two of these.

The first two results that altered the assumed arithmetic properties of α or β were due to G. Ricci and P. Franklin. In 1935 Ricci [24] used Gelfond's approach and introduced a special type of Liouville number into the statement of the Gelfond-Schneider Theorem. Ricci established several theorems with rather complicated statements. Ricci's most easily stated theorem contains the following result.

Theorem 7.1. *Suppose α are β are algebraic numbers where $\alpha \neq 0, 1$ and β is irrational. Suppose further that κ is an irrational number such that for some $\epsilon > 0$*

$$\left|\kappa - \frac{p}{q}\right| < e^{-(\log q)^{2+\epsilon}} \tag{7.1}$$

has infinitely many solutions $p \in \mathbf{Z}, q \in \mathbf{N}$. Then $(\kappa\alpha)^{\beta}$ is transcendental.

Note: The number κ in the statement of Ricci's Theorem is transcendental by Liouville's Theorem, from Chapter 1. An example of such a number κ is a decimal having appropriately increasing sequences of zeros. Consequently, the number $\kappa\alpha$ in the conclusion of this theorem is also transcendental.

In 1937 P. Franklin [6] published a different sort of generalization. Instead of using Liouville numbers, which are well-approximated by a sequence of rational numbers, Franklin considered numbers that are well-approximated by

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algebraic numbers from a fixed number field. In order to appreciate how far transcendental number theory has advanced, let's look at the full statement of one of Franklin's Theorems.

Theorem 7.2. *Let $\{\alpha_i\}, \{\beta_i\},$ and $\{\eta_i\}$ be three sequences of irrational numbers in a fixed number field K, where the conjugates of all of the elements of these sequences are uniformly bounded. Let δ_i be a sequence of integers, becoming infinite, such that $\delta_i\alpha_i, \delta_i\beta_i,$ and $\delta_i\eta_i$ are algebraic integers. Suppose*

$$a = \lim_{n\to\infty} \alpha_i, b = \lim_{n\to\infty} \beta_i \quad \text{and } a^b = \lim_{n\to\infty} \eta_i, \tag{7.2}$$

where $a \neq 0, 1, b \neq 0, 1$. If

$$|a - \alpha_i| + |b - \beta_i| + |a^b - \eta_i| < {\delta_i}^{-(\log \delta_i)^\kappa}, \tag{7.3}$$

then it is impossible to have

$$\kappa > 6.$$

In another direction, within a few years of the solution to the α^β portion of Hilbert's seventh problem, authors began to provide quantitative interpretations of the Gelfond-Schneider Theorem. These quantitative results can take one of two forms depending on how you view the statement: For algebraic numbers α and β, with $\alpha \neq 0, 1$ and β irrational, α^β is transcendental. One is to conclude that for any nonzero integral polynomial $P(x)$, $P(\alpha^\beta) \neq 0$. Another is to conclude that for any algebraic number $\gamma, \alpha^\beta - \gamma \neq 0$. It is these two, related, statements that were first given quantitative versions.

These early results, as with almost all subsequent such results, had very explicit, so fairly complicated, statements. Here we state three of them, due to Gelfond, the first two from 1935 [11] and the third from 1939 [12].

Common hypotheses. *Suppose α and β are algebraic numbers with $\alpha \neq 0, 1$ and β irrational, and suppose that γ is an algebraic number with degree $deg(\gamma)$ and height $H(\gamma)$.*

1. Take $d \in \mathbf{N}$ and $\epsilon > 0$. There exists a constant $c(\alpha, \beta, d, \epsilon)$ such that if $deg(\gamma) \leq d$ and $H(\gamma) > c$ then

$$|\alpha^\beta - \gamma| > H(\gamma)^{-(\log\log H(\gamma))^{5+\epsilon}}.$$

2. Suppose $P(x) \in \mathbf{Z}[x]$ has degree d and height $H(P)$. Then there exists a constant c such that

$$|P(\alpha^\beta)| > e^{-cd^2(d+\log H(P))\log^2(d+\log H(P)+1)\log^{-3}(d+1)}. \tag{7.4}$$

3. Take $d \in \mathbf{N}$ and $\epsilon > 0$. There exists a constant $c(\alpha, \beta, d, \epsilon)$ such that if $deg(\gamma) \leq d$ and $H(\gamma) > c$ then

$$\left|\frac{\log\alpha}{\log\beta} - \gamma\right| > e^{-(\log H(\gamma))^{3+\epsilon}}. \tag{7.5}$$

Each of these results were greatly improved, and generalized, throughout the twentieth century. We do not pursue this history here; instead we refer the reader to [5] or many of the fine survey articles written by Michel Waldschmidt [31]. But we do point out that many of these improvements depended on extensions or improvements of the ideas we have already seen–there was an especially persistent reliance on applications of the pigeonhole principle, mostly in the form of Siegel's Lemma.

In this short chapter we want to illustrate a new method, initiated by M. Laurent in the early 1990s [20],which does not rely on an application of Siegel's Lemma. Laurent's idea is that instead of using Siegel's Lemma to establish the existence of an advantageous function one could directly study the sort of matrix that might underlie an application of Siegel's Lemma. Working directly with this matrix allowed Laurent to give a very precise estimate for the absolute value of a nonzero linear form

$$\Lambda = b_1 \log \alpha + b_2 \log \beta,$$

where b_1 and b_2 are rational integers and α and β are algebraic numbers. Such a result is clearly along the lines of (7.5) if γ is assumed to be rational.

We illustrate Laurent's method not by looking at how he established a lower bound for $|\Lambda|$, above, but by offering a proof of a generalization of (7.4) given by Waldschmidt [32]. (We only look at a corollary of Waldschmidt's result.)

Theorem 7.3. *Let α be a positive real number, $\alpha \neq 1$, and β an irrational real number. Let $\alpha^\beta = e^{\beta \log \alpha}$, where $\log \alpha$ is the real value of the logarithm of α. Then for any sufficiently large rational integer N there exists a polynomial $P \in \mathbf{Z}[X_1^{\pm 1}, X_2^{\pm 1}, Y]$ satisfying*

$$\deg P + \log Ht(P) \leq 5N^{14} \log N,$$

and

$$0 < |P(\alpha, \alpha^\beta, \beta)| < e^{-\frac{1}{3}N^{16}}. \tag{7.6}$$

Before we examine the proof of this theorem, and its use of a determinant in place of a function constructed through an application of the pigeonhole principle, let's see that it has the Gelfond-Schneider Theorem as a corollary.

Corollary 7.4. *Under the hypothesis of the above theorem not all of α, β, and α^β can be algebraic.*

Proof. Suppose that each of the numbers $\alpha, \beta,$ and α^β is algebraic. Let $\alpha_1(= \alpha), \ldots, \alpha_{d_1}$ denote the conjugates of α; $\beta_1(= \beta), \ldots, \beta_{d_2}$ denote the conjugates of β, and $\gamma_1(= \alpha^\beta), \ldots, \gamma_{d_3}$ denote the conjugates of α^β. For simplicity we assume P does not involve negative powers of the variables (if it does we can

multiply through by the variable to the appropriate power). Then

$$M = |P(\alpha, \alpha^\beta, \beta)| \prod_{(i,j,k)\neq(1,1,1)} |P(\alpha_i, \gamma_j, \beta_k)|$$

is a nonzero integer.

Using the estimate of the theorem for $|P(\alpha, \alpha^\beta, \beta)|$, and estimating each of the other terms by the triangle inequality, shows that by taking N sufficiently large the integer M satisfies: $0 < M < 1$.

The proof of Theorem 7.3. The proof of the above theorem does not look like any of the other proofs we have considered but we will see that it has the same essential components (as the following outline indicates).

Outline of the proof

For simplicity, although risking confusing the reader, we let

$$D_1 = N^6 - 1,\ D_2 = \frac{1}{2}(N^2 - 1),\ \text{and}\ K = \frac{1}{2}(N^4 - 1),$$

and we restrict N to be odd so that each of the above parameters is an integer.

Step 1. Instead of looking at an auxiliary function of the form

$$F(z) = \sum_{m=0}^{D_1} \sum_{n=-D_2}^{D_2} a_{mn} z^m \alpha^{nz}$$

at points $k_1 + k_2\beta, -K \leq k_1, k_2 \leq K$, for some parameters $D_1, D_2,$ and K we just consider the collection of functions

$$\phi_{mn} = z^m \alpha^{nz}, 0 \leq m \leq D_1, -D_2 \leq n \leq D_2.$$

We can put any ordering we want on this collection of functions; for clarity we order them lexicographically:

$$\phi_{0,-D_2}, \phi_{0,-D_2+1}, \ldots, \phi_{0,D_2}, \phi_{1,-D_2}, \ldots, \phi_{1,D_2}, \ldots \phi_{D_1,-D_2}, \ldots, \phi_{D_1,D_2}.$$

which we respectively label $\phi_1, \ldots, \phi_L,\ L = (D_1 + 1)(2D_2 + 1)$.

We evaluate each of these functions at the points $k_1 + k_2\beta, -K \leq k_1, k_2 \leq K$; we also order these points lexicographically and then label them according to this ordering: $\zeta_1, \ldots, \zeta_{(2K+1)^2}$.

The remainder of the proof works directly with the matrix which consists of the functions ϕ_{mn} evaluated at the points $k_1 + k_2\beta$. The columns of the matrix are indexed by the ordering of the functions ϕ_λ and the rows are indexed by the ordering of the points ζ_μ. Since we want this to be a square matrix we take

$$L = (D_1 + 1)(2D_2 + 1) = (2K + 1)^2,$$

and, as promised, consider the matrix:

$$\big(\phi_\lambda(\zeta_\mu)\big)_{1\le\lambda,\mu\le L}. \tag{7.7}$$

(The polynomial P in the conclusion of the theorem is the determinant of the matrix above with X_1 replacing α, X_2 replacing α^β, and Y replacing β.)

Step 2. We will denote the determinant of this matrix by Δ. The zeros estimate at the end of Chapter 5 may be applied to conclude that $\Delta \neq 0$.

Step 3. The degree and height of the determinant of the matrix with X_1 replacing α_1, X_2 replacing α^β, and Y replacing β are computed directly from representing this determinant as a sum of products of its entries.

Step 4. The absolute value of the nonzero value from the first step is then estimated from above through an application of the Maximum Modulus Principle.

Details of the proof

Application of the Zeros Estimate. In order to apply the zeros estimate from the end of Chapter 5 to show that Δ does not vanish we need to show that if it did vanish we would have a nonzero exponential polynomial with too many real zeros. If $\Delta = 0$ then its columns are linearly dependent. Using our ordering we can explicitly represent this linear dependency between the columns as:

$$\sum_{\lambda=1}^{L} A_\lambda \phi_\lambda(z)\Big|_{z=\zeta_\mu} = 0,\ 1 \le \mu \le (2K+1)^2. \tag{7.8}$$

This dependency may be translated into function notation:

$$F(z) = \sum_{m=0}^{D_1}\sum_{n=-D_2}^{D_2} a_{mn} z^m \alpha^{nz}\Big|_{(k_1+k_2\beta)} = 0 \tag{7.9}$$

where the coefficients a_{mn} are not all zero. Thus, if $\Delta = 0$ the above function, $F(z)$, vanishes at the L distinct points $k_1 + k_2\beta$.

The application of the zeros estimate is clearer if we rewrite $F(z)$ as

$$F(z) = \sum_{n=-D_2}^{D_2} \left(\sum_{m=0}^{D_1} a_{mn} z^m\right) e^{\omega_n z},$$

where $\omega_n = n \log \alpha$. Then, according to the zeros estimate, $F(z)$ can have at most

$$\underbrace{D_1 + \cdots + D_1}_{(2D_2+1)\text{terms}} + (2D_2+1) - 1 = L - 1,$$

zeros. Thus not all of the values in (7.9) can equal zero, so no such dependency (7.8) can hold. It follows that $\Delta \neq 0$.

Estimating the Degree and Height of Δ: We introduce *monomials* $P_{mn}^{k_1,k_2} = (k_1 + k_2 Y)^m (X_1^{k_1} X_2^{k_2})^n$, where we italicized the word monomials since they can have negative degrees, so that

$$P_{mn}^{k_1,k_2}(\alpha, \alpha^\beta, \beta) = \phi_{mn}(k_1 + k_2\beta).$$

If we use the lexicographical ordering on the subscripts of these polynomials to index the columns of a matrix, and use the lexicographical ordering on the superscripts of these polynomials to index the rows of a matrix, then the matrix we considered above,

$$\big(\phi_\lambda(\zeta_\mu)\big)_{1\le\lambda,\mu\le L}$$

is the same as the matrix

$$\big(P_{mn}^{k_1,k_2}\big) \tag{7.10}$$

evaluated at $X_1 = \alpha, X_2 = \alpha^\beta$, and $Y = \beta$. Notice that we have the following easy estimates:

$$\deg P_{mn}^{k_1,k_2} \le (D_1 + 2D_2K) \text{ and } H(P_{mn}^{k_1,k_2}) \le (2K)^{D_1}. \tag{7.11}$$

The determinant of the matrix (7.10) is a polynomial expression in $X_1^\pm, X_2^\pm, Y$, which we denote by P. In order to estimate the degree and height of P we note that it is a signed sum of terms each of which is a product of L polynomials, one from each row and each column of the matrix (7.10). From this observation we have the estimate

$$\deg P \le L(D_1 + 2D_2K).$$

Estimating the height of P is a more complicated matter. Indeed it is easier to do if we introduce yet-another new concept: For a polynomial Q we define the *length* of Q, denoted $L(Q)$, to be the sum of the absolute values of its coefficients. The reason the length of a polynomial is such a useful concept is because the following simple relationships hold: For any polynomials P and Q,

$$L(P+Q) \le L(P) + L(Q) \text{ and } L(PQ) \le L(P)L(Q).$$

Notice that each of the monomials $P_{mn}^{k_1,k_2}$ has length at most $(2K)^{D_1}$. So from the above characterization of the determinant, using the above two identities about the lengths of the sum and product of two polynomials, we have

$$L(P) \le L!(\max\{L(P_{mn}^{k_1,k_2})\})^L \le L!(2K)^{LD_1}.$$

But, clearly $H(P) \le L(P)$.

An Upper Bound for Δ: So far we have established the lower bound $|\Delta| = |P(\alpha, \alpha^\beta, \beta)| > 0$. The upper bound is easier; it can be deduced from the following lemma.

Lemma 7.5. *Suppose that $f_1(z), f_2(z), \dots, f_L(z)$ are functions analytic in a set containing the disc $D = \{z : |z| \le R\}$. Suppose that $\zeta_1, \dots, \zeta_L$ all have*

absolute value at most r, with $0 < r < R$. Then the determinant

$$\Delta = \begin{pmatrix} f_1(\zeta_1) & \cdots & f_L(\zeta_1) \\ \vdots & \ddots & \vdots \\ f_1(\zeta_L) & \cdots & f_L(\zeta_L) \end{pmatrix}$$

satisfies

$$|\Delta| \leq \left(\frac{R}{r}\right)^{-L(L-1)/2} L! \prod_{\lambda=1}^{L} \max_{|\zeta|=R}\{|f_\lambda(\zeta)|\}.$$

Sketch of proof. The idea of this proof is to introduce a new variable z and consider the function:

$$h(z) = \det\left(f_\lambda(\zeta_j z)\right)$$

and show that $h(z)$ has a zero at $z = 0$ to order at least $L(L-1)/2$. The key to this is to replace each of the functions $f_\lambda(\zeta_j z)$ by its Taylor series expansion at the origin and apply the multi-linearity of the determinant. This allows us to reduce the problem to the case of functions $f_\lambda(z) = z^{n_\lambda}, 1 \leq \lambda \leq L$, where each n_λ is a non-negative integer. In this simple case we have

$$h(z) = z^{n_1+n_2+\cdots+n_L} \det\left(\zeta_j^{n_\lambda}\right).$$

If $h(z)$ is not identically zero then the Vandermonde determinant of the matrix $\left(\zeta_\lambda^{n_\lambda}\right)$ is nonzero. Thus the non-negative integers $n_1, \ldots, n_L$ are pairwise distinct. Then the sum $n_1 + \cdots + n_L$ is at least $0 + 1 + \cdots + (L-1)$ which equals $L(L-1)/2$. This implies that the order of vanishing of $h(z)$ at the origin is at least $L(L-1)/2$.

We now use this zero at $z = 0$ of order at least $L(L-1)/2$ to obtain the desired upper bound. The function

$$G(z) = \frac{h(z)}{z^{L(L-1)/2}}$$

is analytic in the disc $|z| \leq R$, and since $r < R$ we have $|G|_r \leq |G|_R$. By the Maximum Modulus Principle we also have

$$|G|_r = r^{-L(L-1)/2}|h|_r \quad \text{and } |G|_R = R^{-L(L-1)/2}|h|_R.$$

Thus:

$$|h|_r \leq \left(\frac{R}{r}\right)^{-L(L-1)/2}|h|_R.$$

If we now imagine using this inequality with 1 replacing r and R/r replacing R we obtain

$$|h|_1 \leq \left(\frac{R}{r}\right)^{-L(L-1)/2}|h|_{R/r}.$$

The reason we imagined using the two radii of 1 and R/r is because when we have $|z| \leq R/r$ we have $|\zeta_j z| \leq R$.

Expanding the determinant Δ we get $L!$ terms each being plus or minus a sum of elements, one taken from each column and one taken from each row. So we obtain

$$|\Delta| = |h(1)| \le |h|_1 \le \left(\frac{R}{r}\right)^{-L(L-1)/2} \times L! \prod_{\lambda=1}^{L} |f_\lambda|_R,$$

which establishes the lemma.

We apply this lemma with $r = K(1 + |\beta|)$ and $R = er$ to obtain the upper bound on $|\Delta|$ of Theorem 7.3, concluding its proof.

Exercises

1. Prove that the number κ in the statement of Theorem 7.1 is transcendental.

2. Suppose P and Q are polynomials in one variable. Establish the following inequalities involving their lengths $L(P)$ and $L(Q)$:

a) $L(P) + L(Q) \le L(P) + L(Q)$

b) $L(P)L(Q) \le L(P)L(Q)$

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Index

algebraic number
  conjugates, 19
  denominator, 19
  height, 50
  norm, 19

Boehle, K., 21
  Theorem, 30

Dirichlet
  box principle, 33

Euler, L., 2
  conjecture, 2, 5
  irrational number (meaning), 2
  ratio of logarithms, 2
  transcendental numbers, 3

Fourier, J.
  irrationality of e, 3, 6
Franklin, P.
  Theorem, 74
Fukasawa, S.
  products of Gaussian integers, 28
function
  elliptic, 67
  growth and number of zeros, 59, 65
  order of growth
    definition, 65, 72
  transcendental, 61
  e^z, 23
  transcendental, definition, 4
  Weierstrass $\wp-$function, 62
  Zeta, 2
functions, algebraically independent, 30, 62

Gaussian integers
  definition, 22
  Gelfond's ordering, 22
    size of nth Gaussian integer, 24, 31
  in a disc, 24, 31
  unique factorization domain, 26
Gelfond, A. O., 5
  e^π is transcendental, 21
  quantitative results, 74
  Siegel's Lemma with inequalities, 37
  solution to Hilbert's Seventh Problem, 34
Gelfond-Schneider Theorem, 58, 61, 64, 73, 75
  and functions, 62
  and Schneider-Lang Theorem, 65
  elliptic analogue, 71
  quantitative versions, 74, 75

Hermite, C., 3
  transcendence of e, 4, 15

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Hermite-Lindemann Theorem, 62, 64
  elliptic analogue, 71
  transcendence of e^{α}, 18
Hilbert, D., vii, 1
  α^{β}, 5
  conjecture of (three versions), 5
  Seventh Problem, 4
  twenty-three problems, 1
Hille, E., 34
Hurwitz, A., 10, 18

International Congress of Mathematicians, 1
interpolation
  Lagrange formula, 29
  Newton Series, 22
irrational number
  $2^{\sqrt{2}}$, 3

Kuzmin, R. O., 21
  $2^{\sqrt{2}}$ is transcendental, 29

Lang, S., 65
Laurent, M.
  and Siegel's Lemma, 75
Lindemann's Theorem, 22
Lindemann, F.
  transcendence of π, 4
Lindemann-Weierstrass Theorem, 4, 20, 61
Liouville, Joseph, 3
  theorem, 3
    corollary, 11
    proof, 11

pigeonhole principle, 33, 36, 45
Poincaré, H., 1
Polya, G.
  zeros of an exponential polynomial, 58
    and Laurent's method, 77
    and Schneider's proof, 59
polynomial
  height, 50
  length, 78, 80
power series
  e^{z}, 14
  e, 3, 6

Ricci, G.
  Theorem, 73

Schneider, Th., 5
  Siegel's Lemma, 48
  solution to Hilbert's Seventh Problem, 45
Schneider-Lang Theorem
  preliminary versions, 64
  statement, 65
series
  Newton interpolation for $e^{\pi z}$, 22
Siegel, C. L., 45
  transcendence of a period of $\wp-$function, 69
Six Exponentials Theorem, 62

theorem
  Boehle (1930), 30
  Fourier (1815), 6
  Franklin (1937), 74
  Gelfond
    e^{π} (1929), 21
    quantitative Gelfond-Schneider (1935, 1939), 74
    solution to Hilbert's Seventh Problem (1934), 34
  Hermite (1873), 15
  Hermite-Lindemann (1882), 18
  Kuzmin (1930), 29
  Lindemann-Weierstrass (1885), 4
  Liouville (1840), 3
  Ricci (1935), 73
  Schneider
    algebraic values of algebraically independent functions (1949), 65
    period of an elliptic function (1934), 69
    solution to Hilbert's Seventh Problem (1934), 45
  Schneider-Lang (1966), 65
  Six Exponentials (1966), 62

Waldschmidt (2000), 75
Thue, A., 45
transcendental number
α^β, 5
$\cos(\alpha)$ and $\sin(\alpha)$, 11
definition, 1
e, 4
proof, 15
e^π, 21
e^α, 18
$\frac{\log\alpha}{\log\beta}$, 34
Liouville, 3, 11
nonzero period of $\wp(z)$, 69
π, 4
values of Gamma function, 70
triangle, isosceles
and Hilbert's Seventh Problem, 11

Vandermonde determinant, 35, 79
Gelfond's use thereof, 44
Schneider's use thereof, 55, 57

Waldschmidt, M., 75
quantitative Gelfond-Schneider Theorem, 75
Weierstrass, K.
$\wp-$function, 62
Laurent series, 68
properties, 67

Zeta function, 2